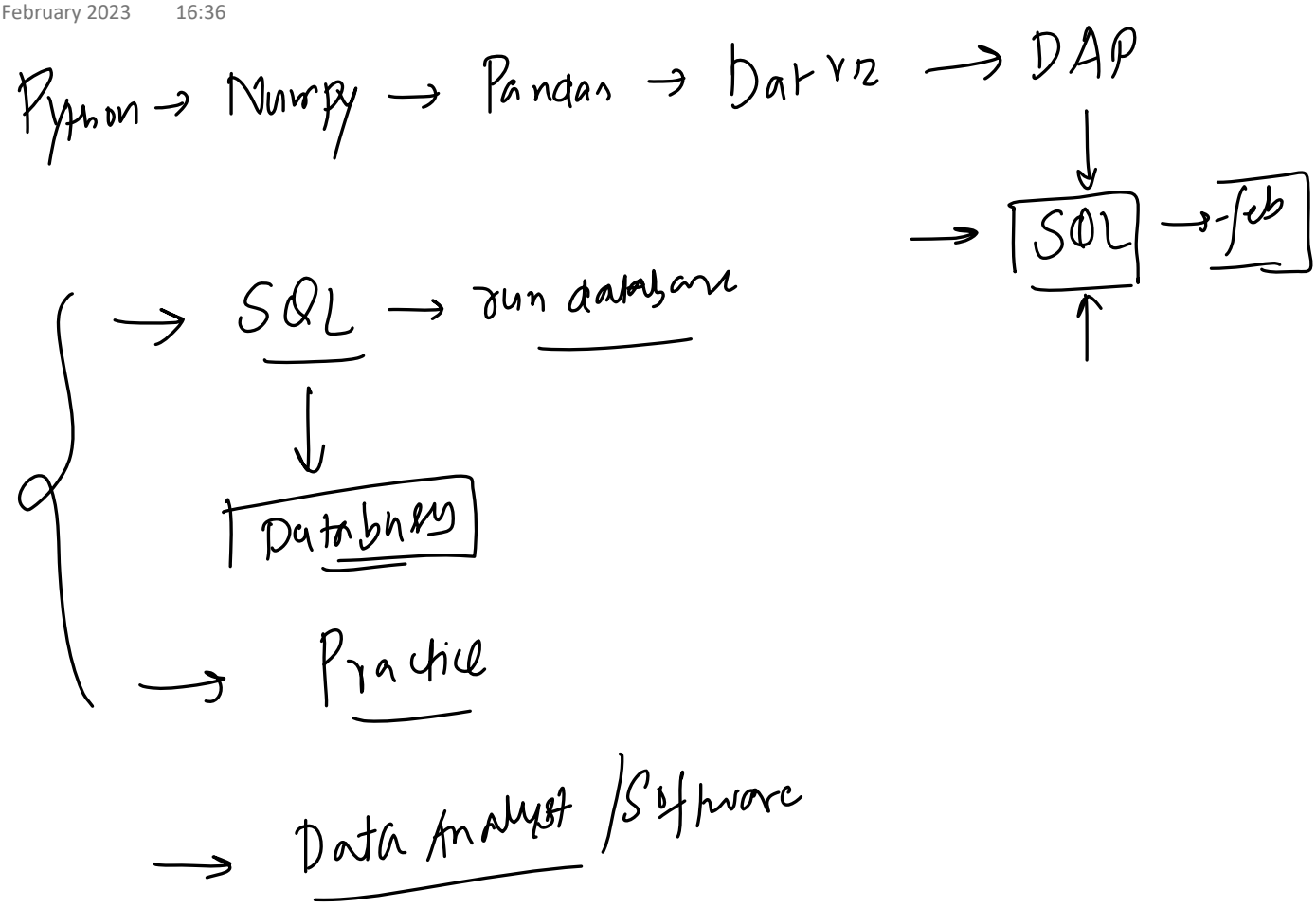
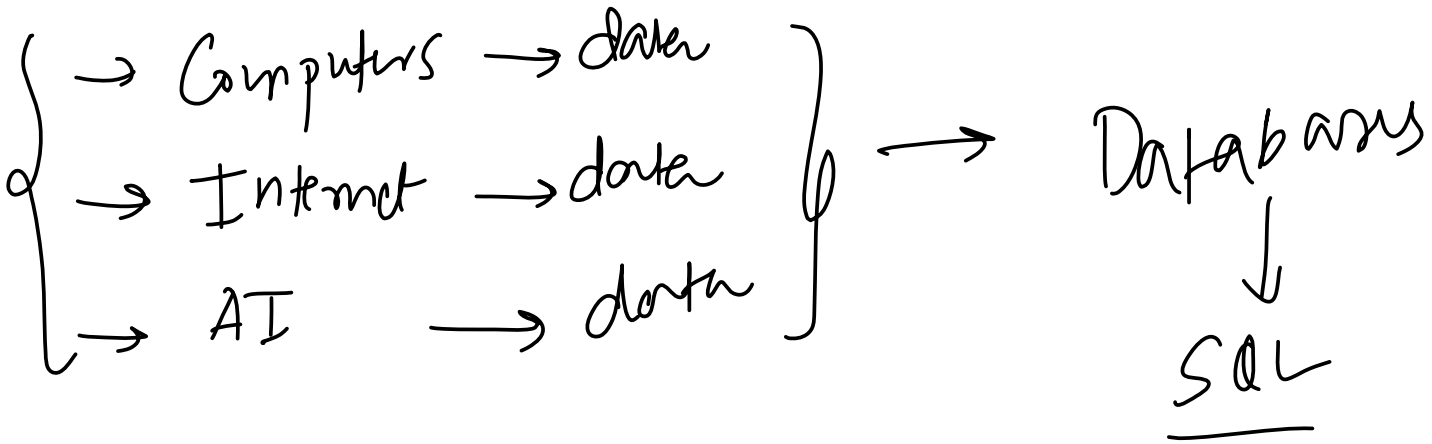
1. Before starting

06 February 2023 16:36



2. Importance of Data

06 February 2023 16:36



3. What are Databases?

06 February 2023 16:37

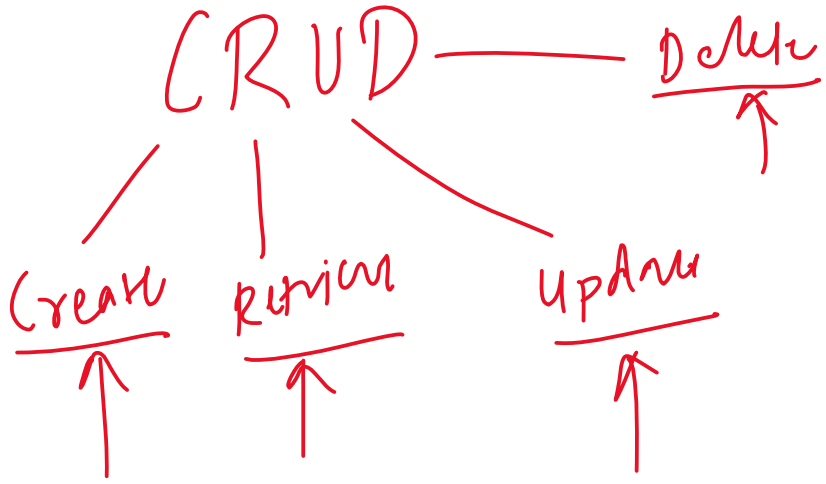
A Database is a shared collection of logically related data and description of these data, designed to meet the information needs of an organization

Data Storage: A database is used to store large amounts of structured data, making it easily accessible, searchable, and retrievable.

Data Analysis: A database can be used to perform complex data analysis, generate reports, and provide insights into the data.

Record Keeping: A database is often used to keep track of important records, such as financial transactions, customer information, and inventory levels.

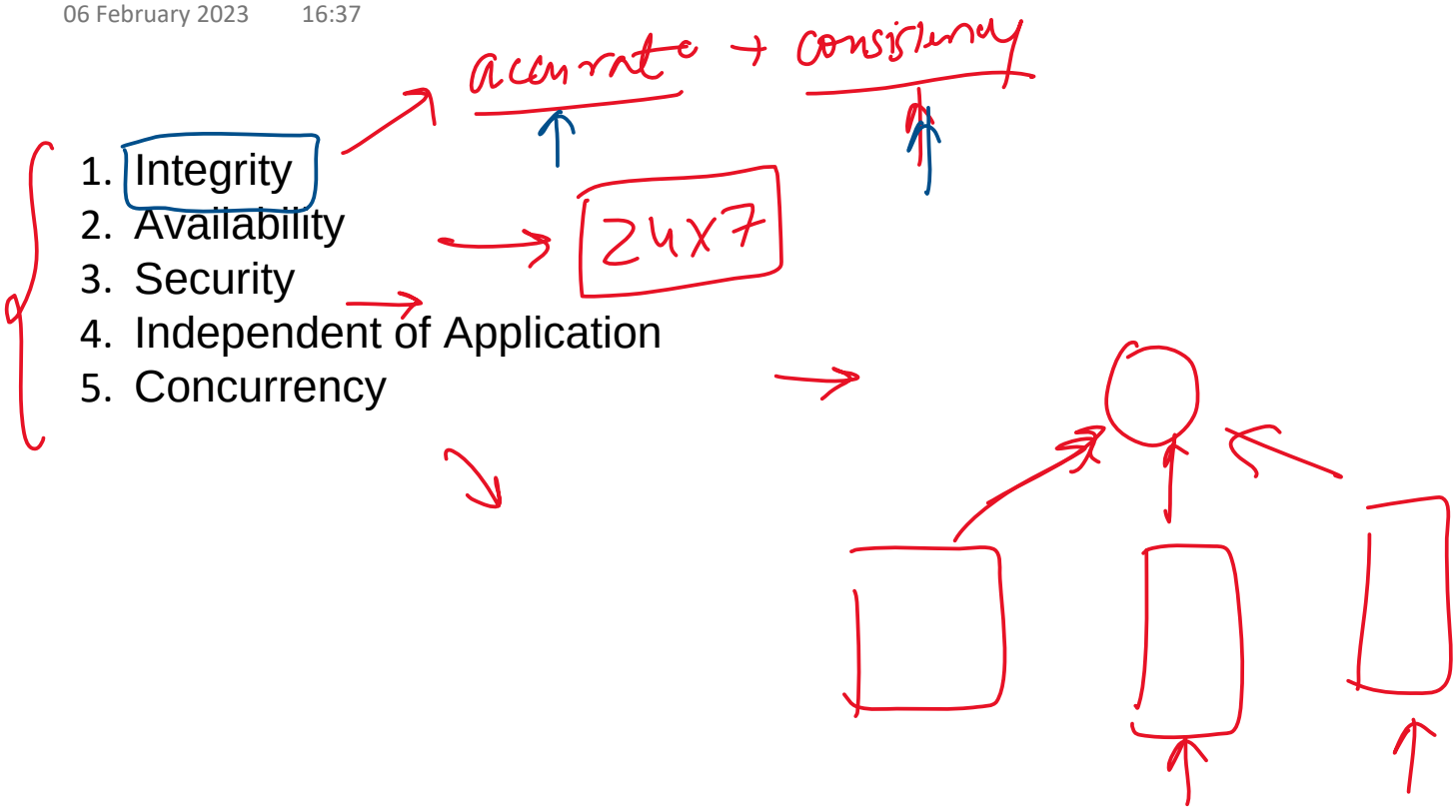
Web Applications: Databases are an essential component of many web applications, providing dynamic content and user management.



4. Properties of an Ideal Database

06 February 2023 16:37

1. Integrity
2. Availability
3. Security
4. Independent of Application
5. Concurrency

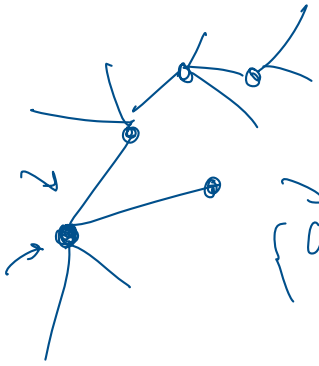


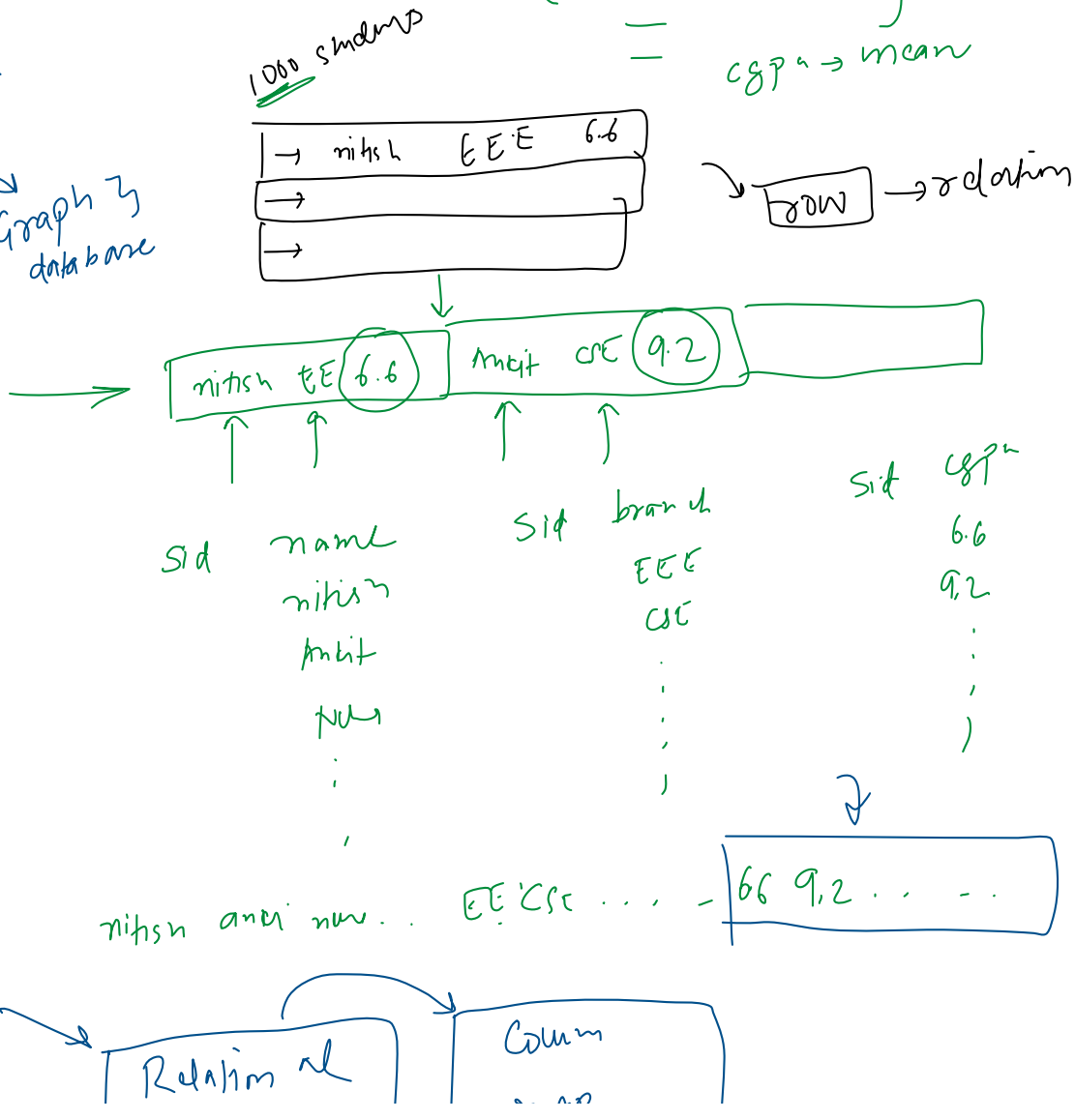
5. Types of Databases

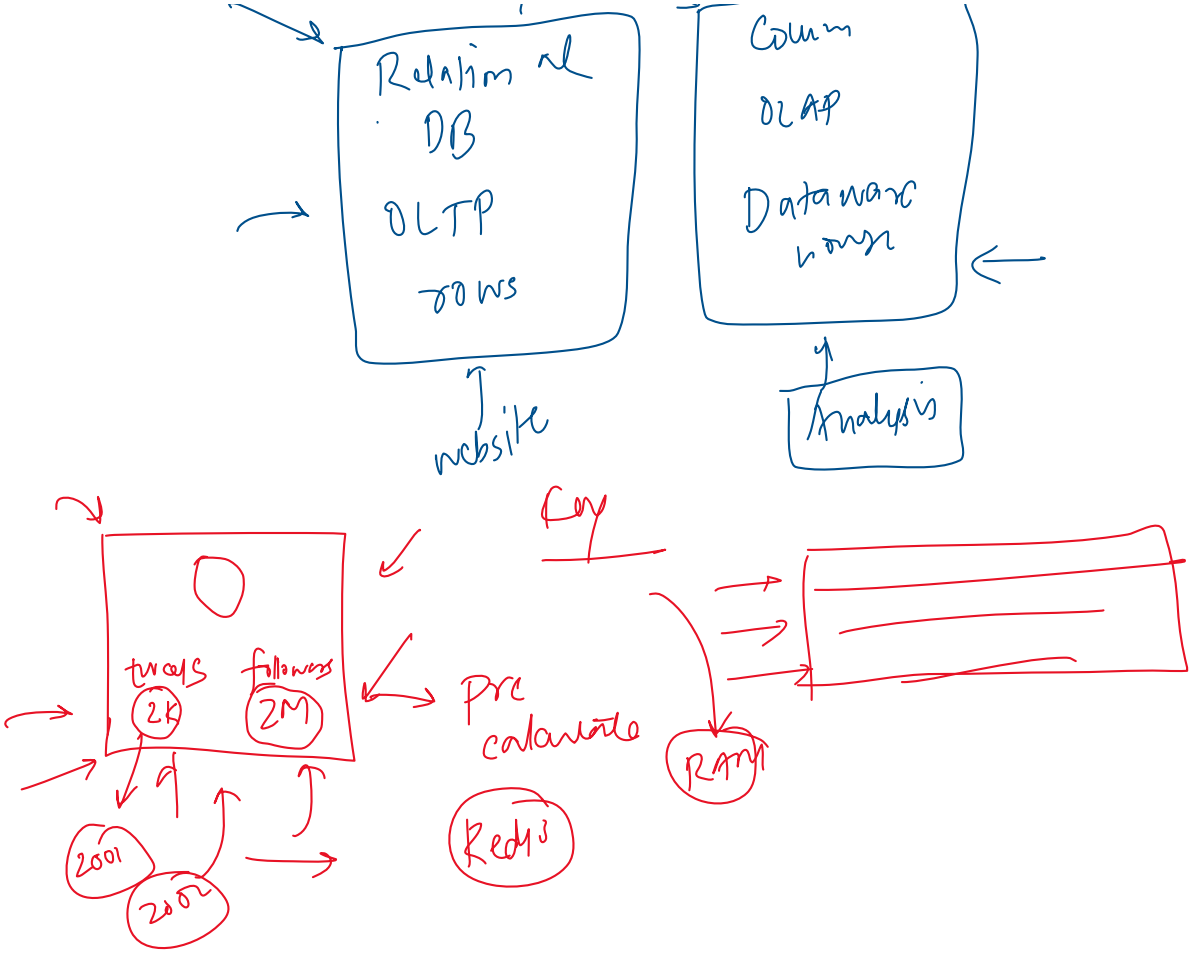
06 February 2023 16:42

1. Relational Databases -
   Also known as SQL databases, these databases use a relational model to organize data into tables with rows and columns.
2. NoSQL Databases -
   These databases are designed to handle large amounts of unstructured or semi-structured data, such as documents, images, or videos. (MongoDB)
3. Column Databases -
   These databases store data in columns rather than rows, making them well-suited for data warehousing and analytical applications. (Amazon Redshift, Google BigQuery)
4. Graph Databases -
   These databases are used to store and query graph-structured data, such as social network connections or recommendation systems. (Neo4j, Amazon Neptune)
5. Key-value databases -
   These databases store data as a collection of keys and values, making them well-suited for caching and simple data storage needs (Redis and Amazon DynamoDB)

Which one should you use?



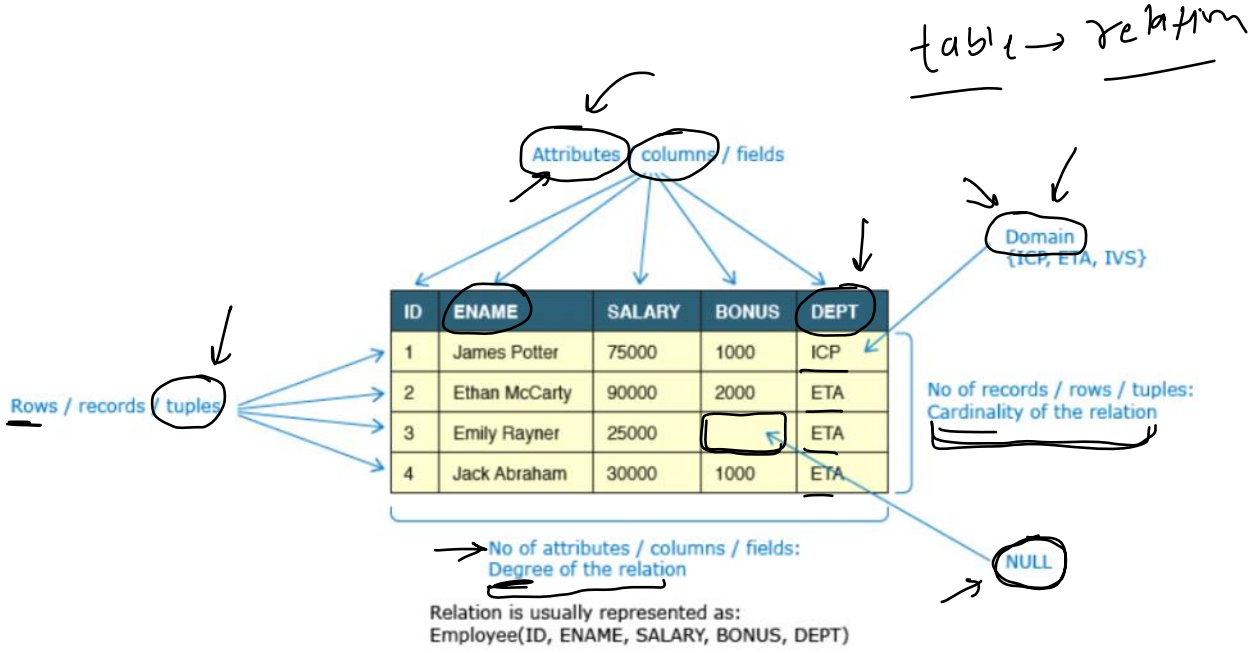



Relational
DB
OLTP
rows
website
Column
OLAP
Data warehouse
Analysis
twoqls
followers
2K
2M
2001
200
Pre calculate
Redis
key
RAM

6. Relational Databases

06 February 2023 16:42

Also known as SQL databases, these databases use a relational model to organize data into tables with rows and columns.

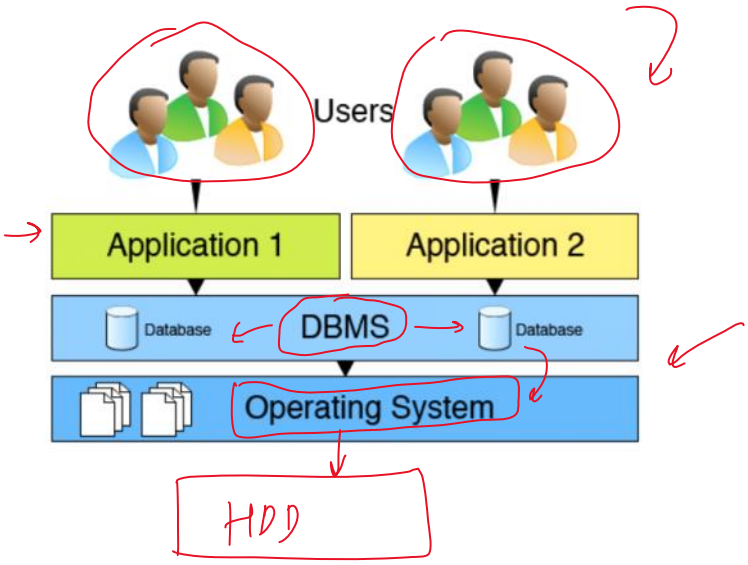


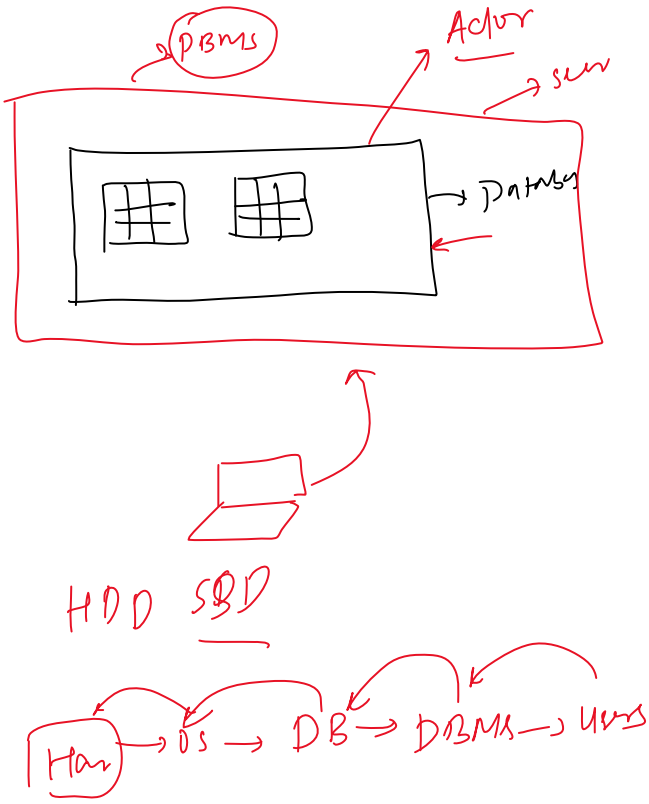
ID	ENAME	SALARY	BONUS	DEPT
1	James Potter	75000	1000	ICP
2	Ethan McCarty	90000	2000	ETA
3	Emily Rayner	25000		ETA
4	Jack Abraham	30000	1000	ETA

7. What is a DBMS

06 February 2023 16:41

A database management system (DBMS) is a software system that provides the interfaces and tools needed to store, organize, and manage data in a database. A DBMS acts as an intermediary between the database and the applications or users that access the data stored in the database.





8. Core Functionalities of a DBMS

06 February 2023 16:41

Functions of DBMS

Data Management - Store, retrieve and modify data

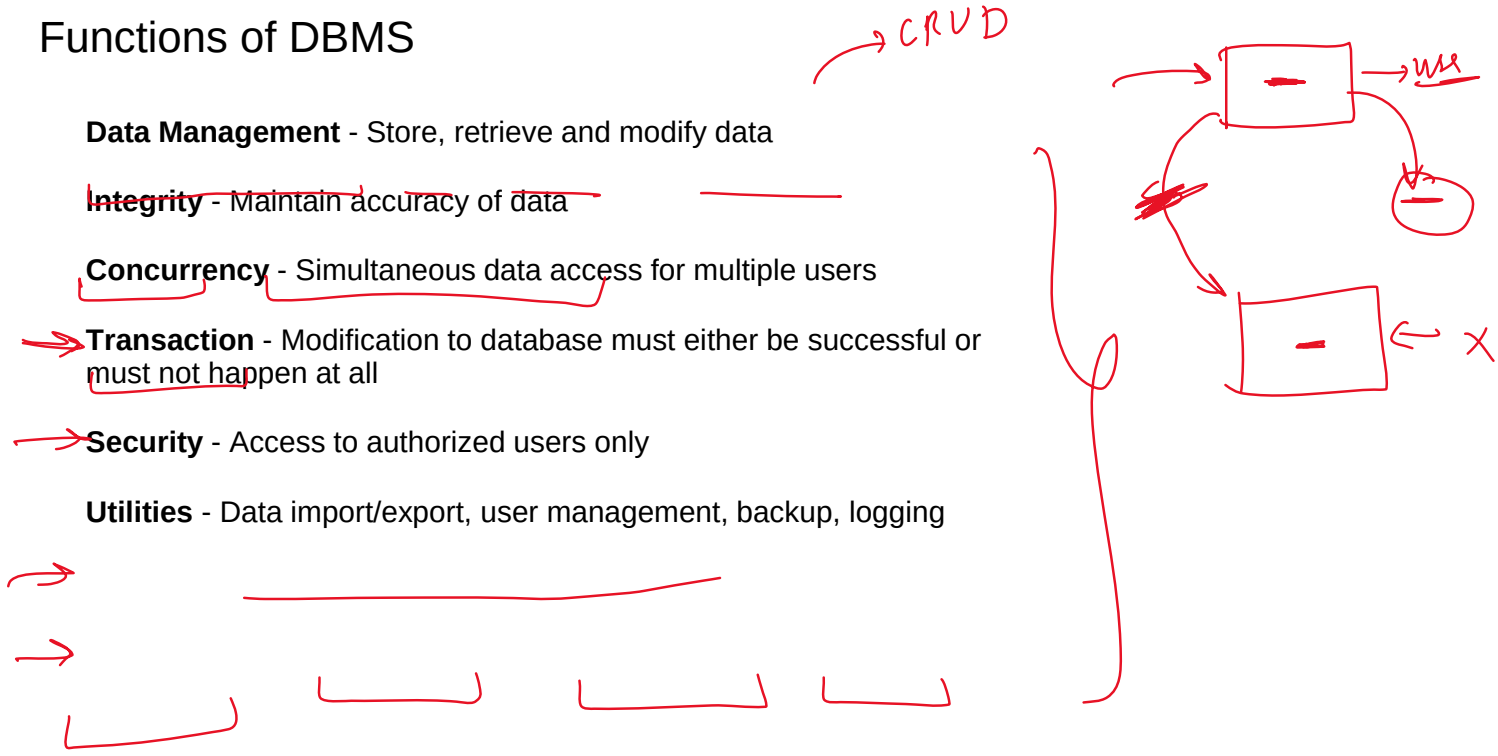
Integrity - Maintain accuracy of data

Concurrency - Simultaneous data access for multiple users

Transaction - Modification to database must either be successful or must not happen at all

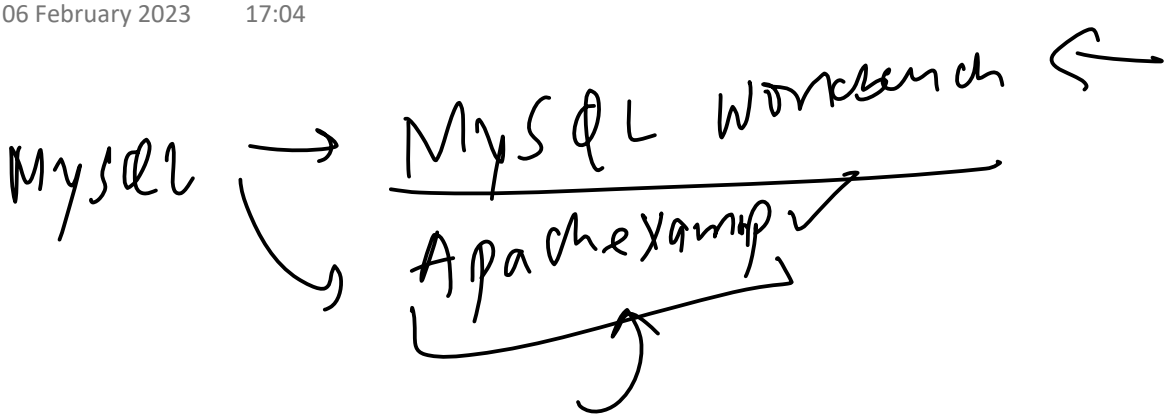
Security - Access to authorized users only

Utilities - Data import/export, user management, backup, logging



9. Practical

06 February 2023 17:04



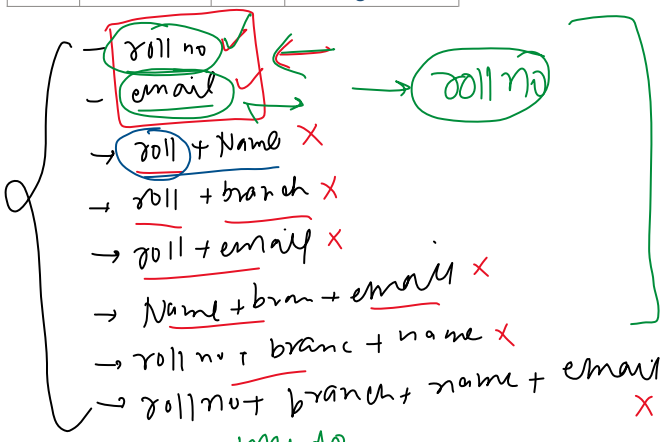
10. Database Keys

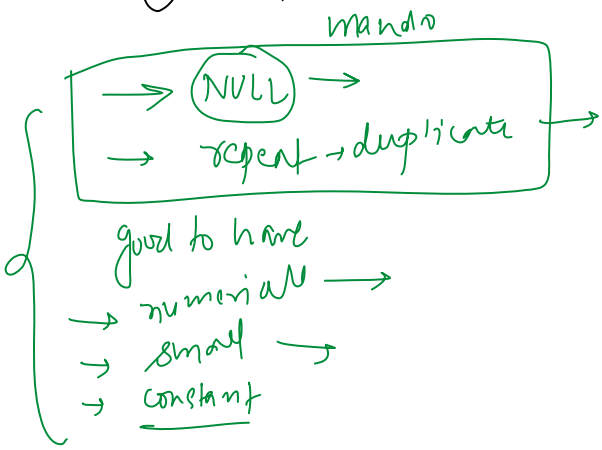
06 February 2023 17:07

A key in a database is an attribute or a set of attributes that uniquely identifies a tuple (row) in a table. Keys play a crucial role in ensuring the integrity and reliability of a database by enforcing unique constraints on the data and establishing relationships between tables.

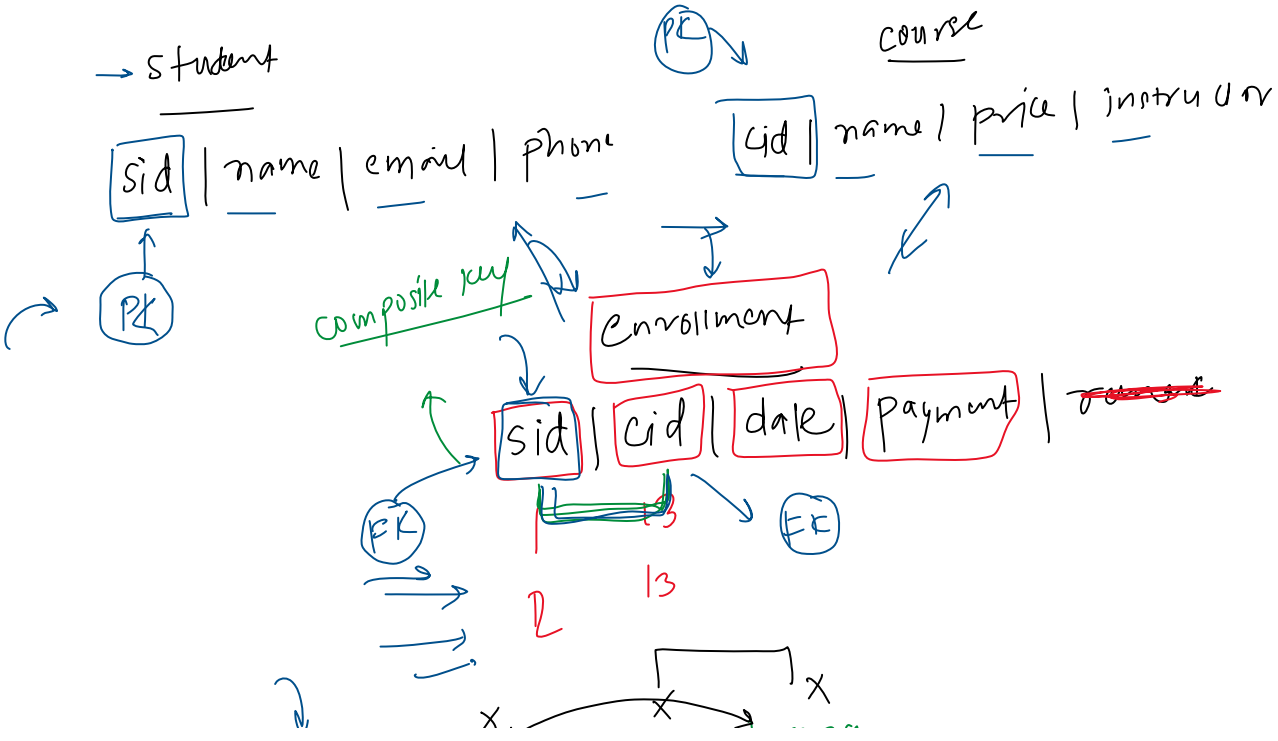
1. **Super Key** -
   A Super key is a combination of columns that uniquely identifies any row within a relational database management system (RDBMS) table
2. **Candidate key** -
   A candidate key is a minimal Super key, meaning it has no redundant attributes. In other words, it's the smallest set of attributes that can be used to uniquely identify a tuple (row) in the table
3. **Primary Key** -
   A primary key is a unique identifier for each tuple in a table. There can only be one primary key in a table, and it cannot contain null values.
4. **Alternate Key** -
   An alternate key is a candidate key that is not used as the primary key.
5. **Composite Key** -
   A composite key is a primary key that is made up of two or more attributes. Composite keys are used when a single attribute is not sufficient to uniquely identify a tuple in a table.
6. **Surrogate Key** -
7. **Foreign Key** -
   A foreign key is a primary key from one table that is used to establish a relationship with another table.

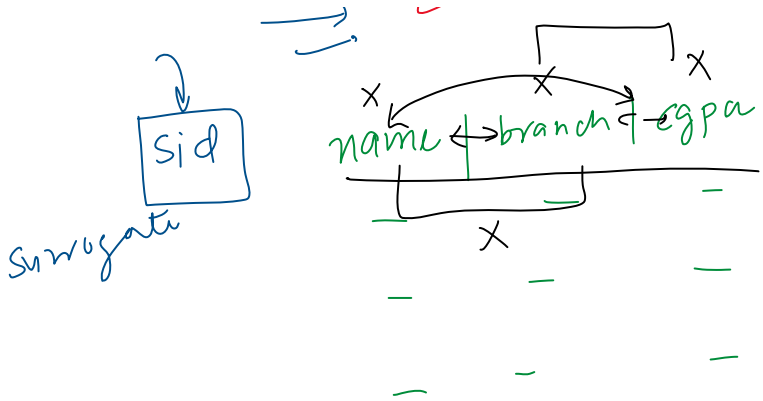
Roll no	Name	Branch	Email
1	Nitish Singh	CSE	nitish@gmail.com
2	Ankit Sharma	EEE	ankit@gmail.com
3	Neha Verma	ME	neha@gmail.com





CK − PK = AK

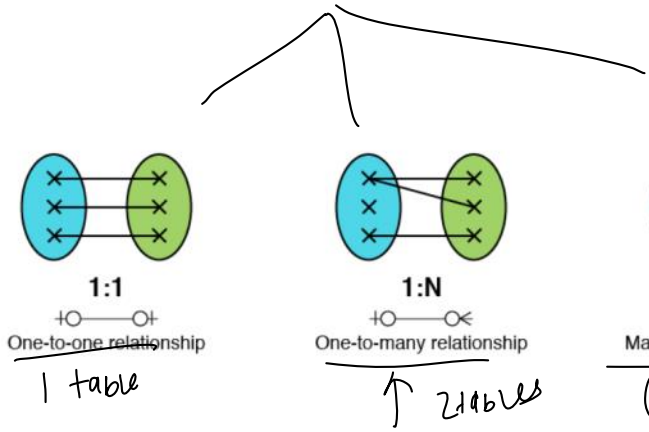



sid
Surrogate
name
branch
cgpa

11. Cardinality of Relationships

06 February 2023 16:43

Cardinality in database relationships refers to the number of occurrences of an entity in a relationship with another entity. Cardinality defines the number of instances of one entity that can be associated with a single instance of the related entity.



Examples

1. Person -> Driving License Number
2. Student -> college branch
3. Restaurants -> orders
4. Restaurants -> menu
5. Students -> courses

12. Drawbacks of Databases

06 February 2023 16:39

Complexity: Setting up and maintaining a database can be complex and time-consuming, especially for large and complex systems.

Cost: The cost of setting up and maintaining a database, including hardware, software, and personnel, can be high.

Scalability: As the amount of data stored in a database grows, it can become more difficult to manage, leading to performance and scalability issues.

Data Integrity: Ensuring the accuracy and consistency of data stored in a database can be a challenge, especially when multiple users are updating the data simultaneously.

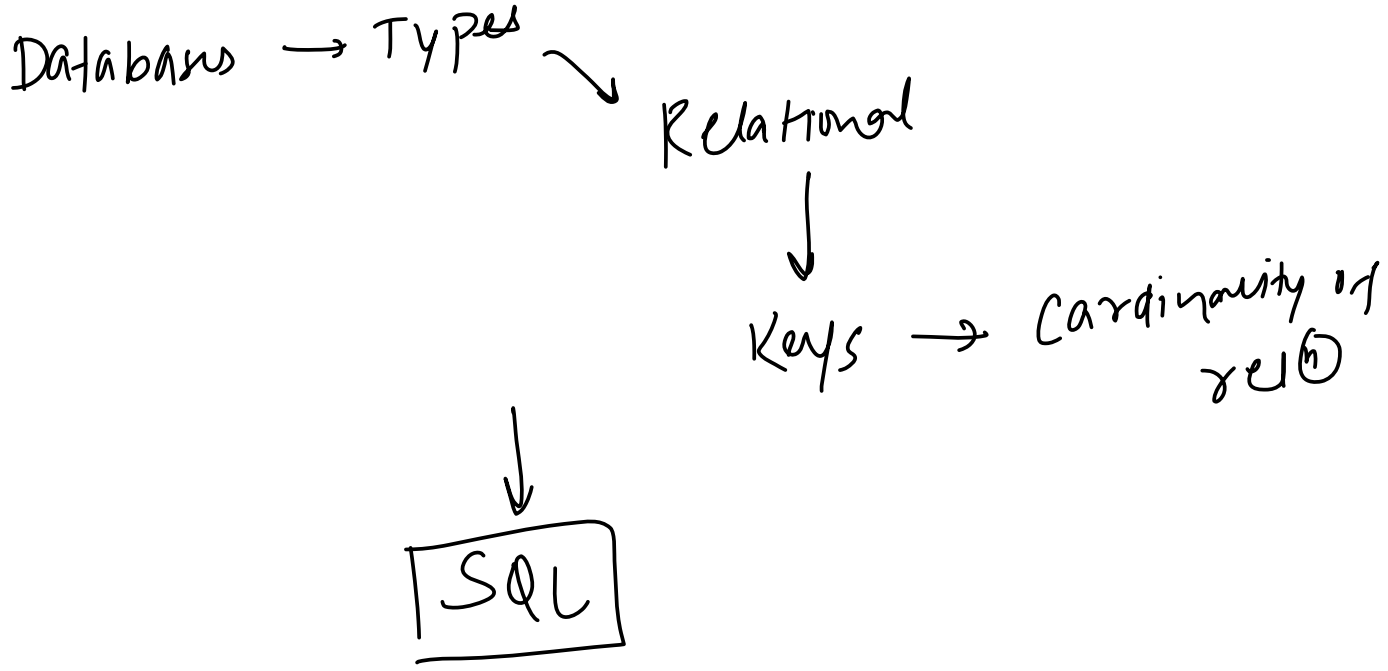
Security: Securing a database from unauthorized access and protecting sensitive information can be difficult, especially with the increasing threat of cyber attacks.

Data Migration: Moving data from one database to another or upgrading to a new database can be a complex and time-consuming process.

Flexibility: The structure of a database is often rigid and inflexible, making it difficult to adapt to changing requirements or to accommodate new types of data.

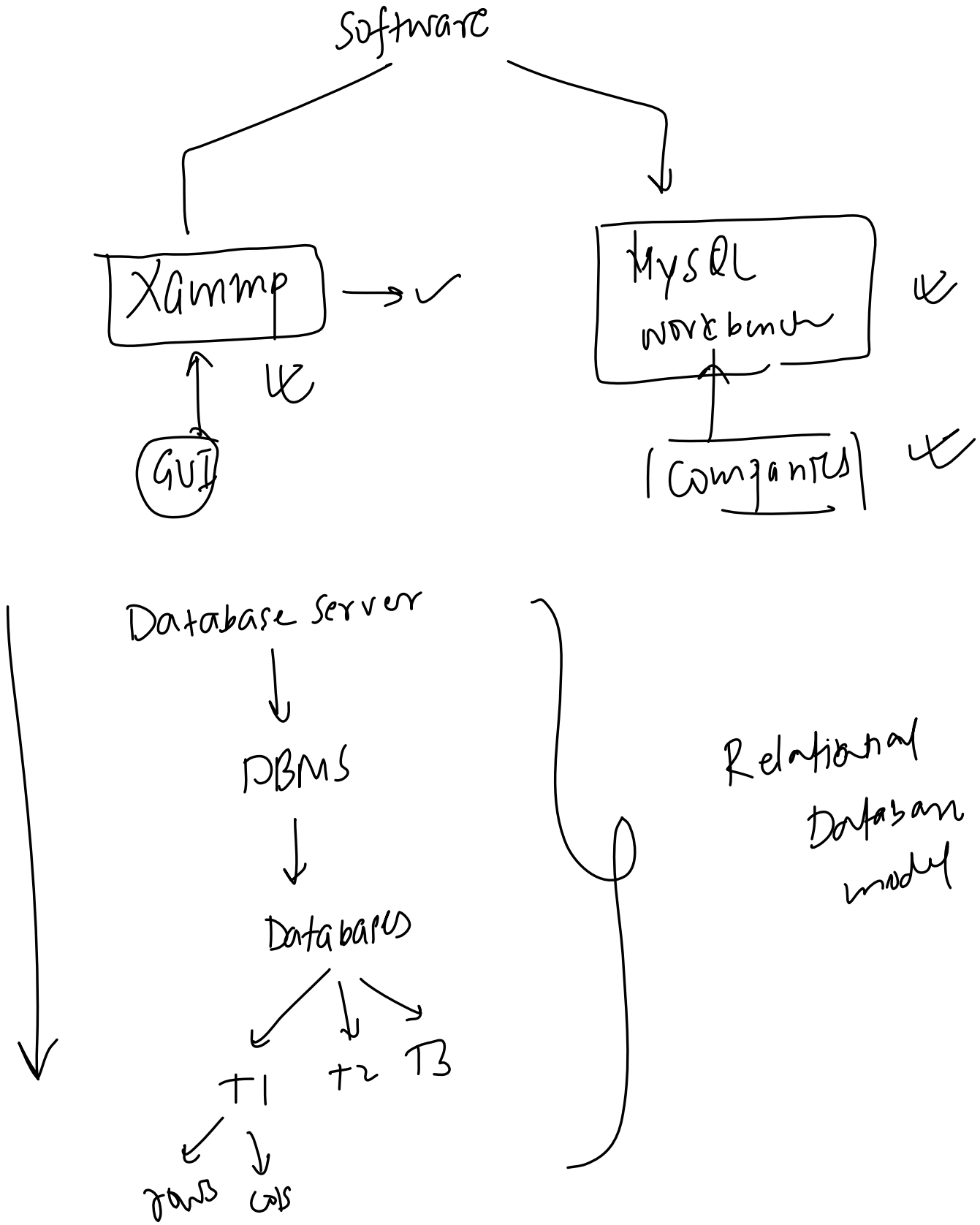
Till now?

08 February 2023 14:56



Installation

08 February 2023 16:05



What is SQL?

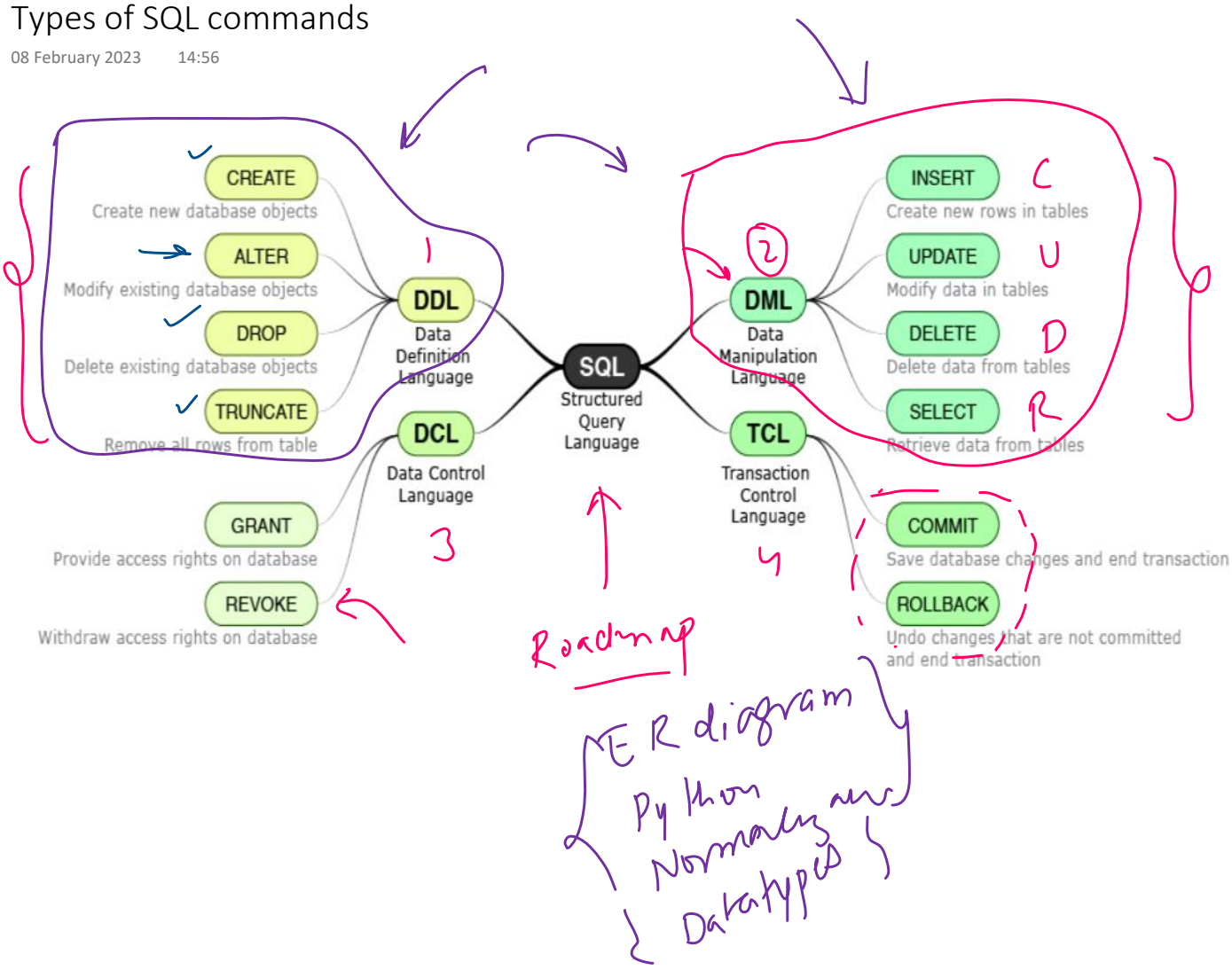
08 February 2023 14:56

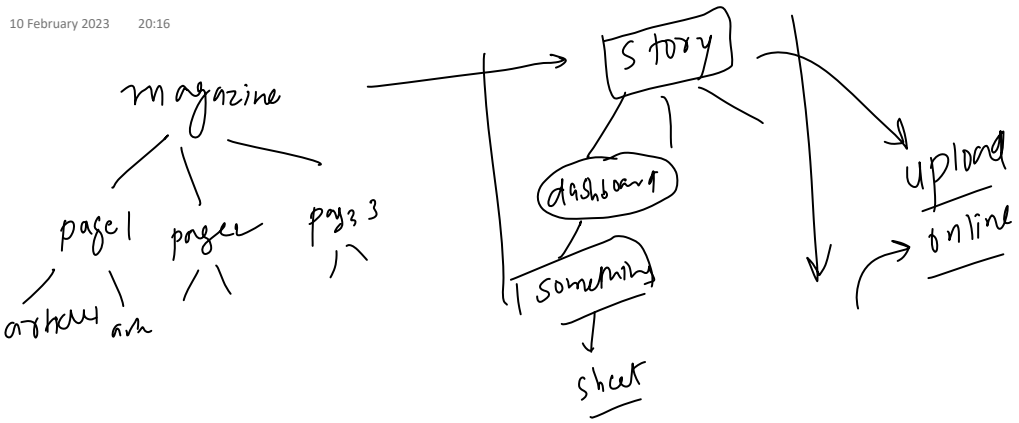
CRUD

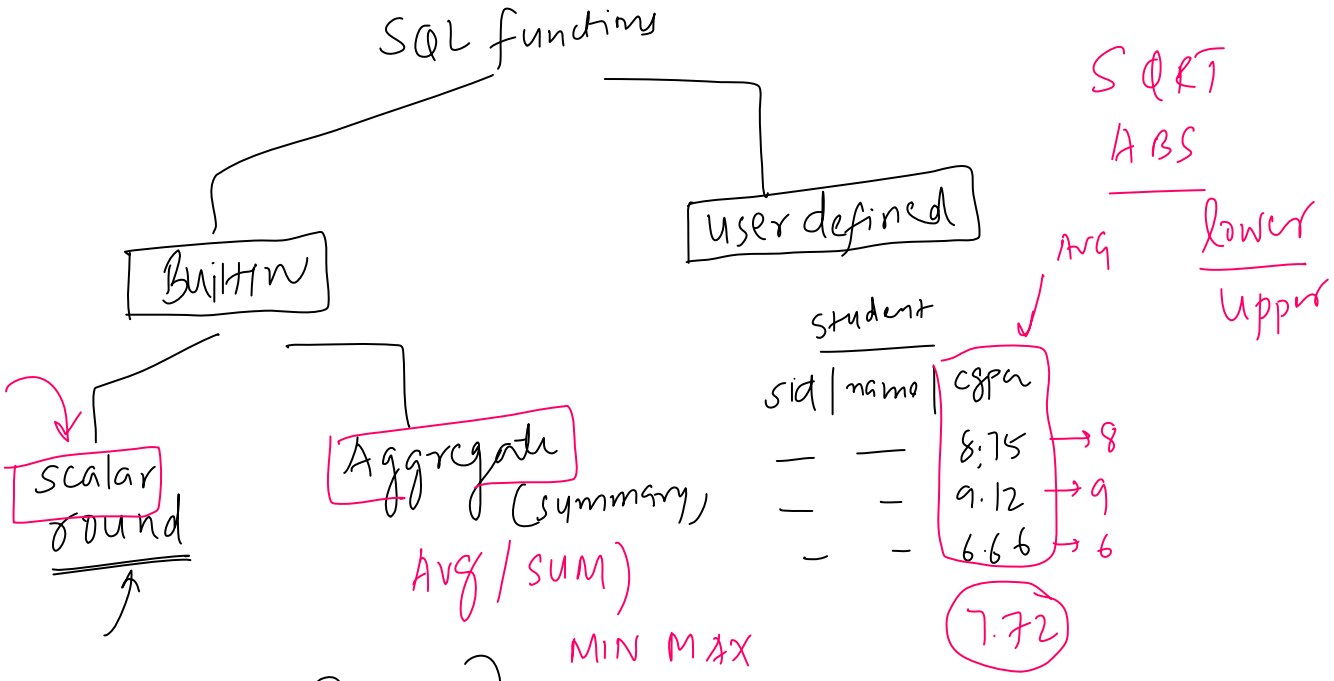
SQL (Structured Query Language) is a programming language used for managing and manipulating data in relational databases. It allows you to insert, update, retrieve, and delete data in a database. It is widely used for data management in many applications, websites, and businesses. In simple terms, SQL is used to communicate with and control databases.

Types of SQL commands

08 February 2023 14:56







name	age	cgpa
N	21	8
A	22	9
R	23	6

sort (Asc) →

R	23	6
N	21	8
A	22	9

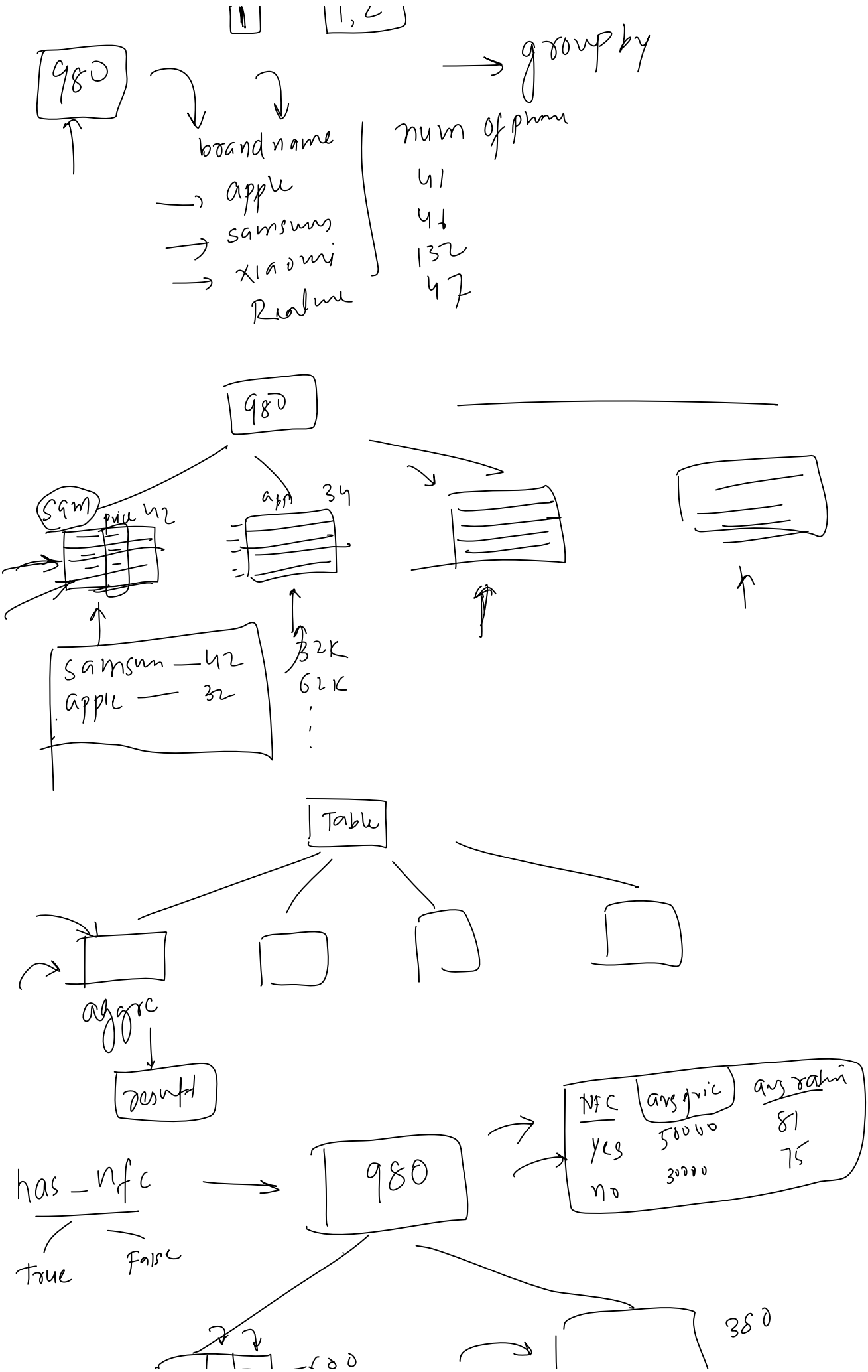
Smart costly phone → price sorting → DESC

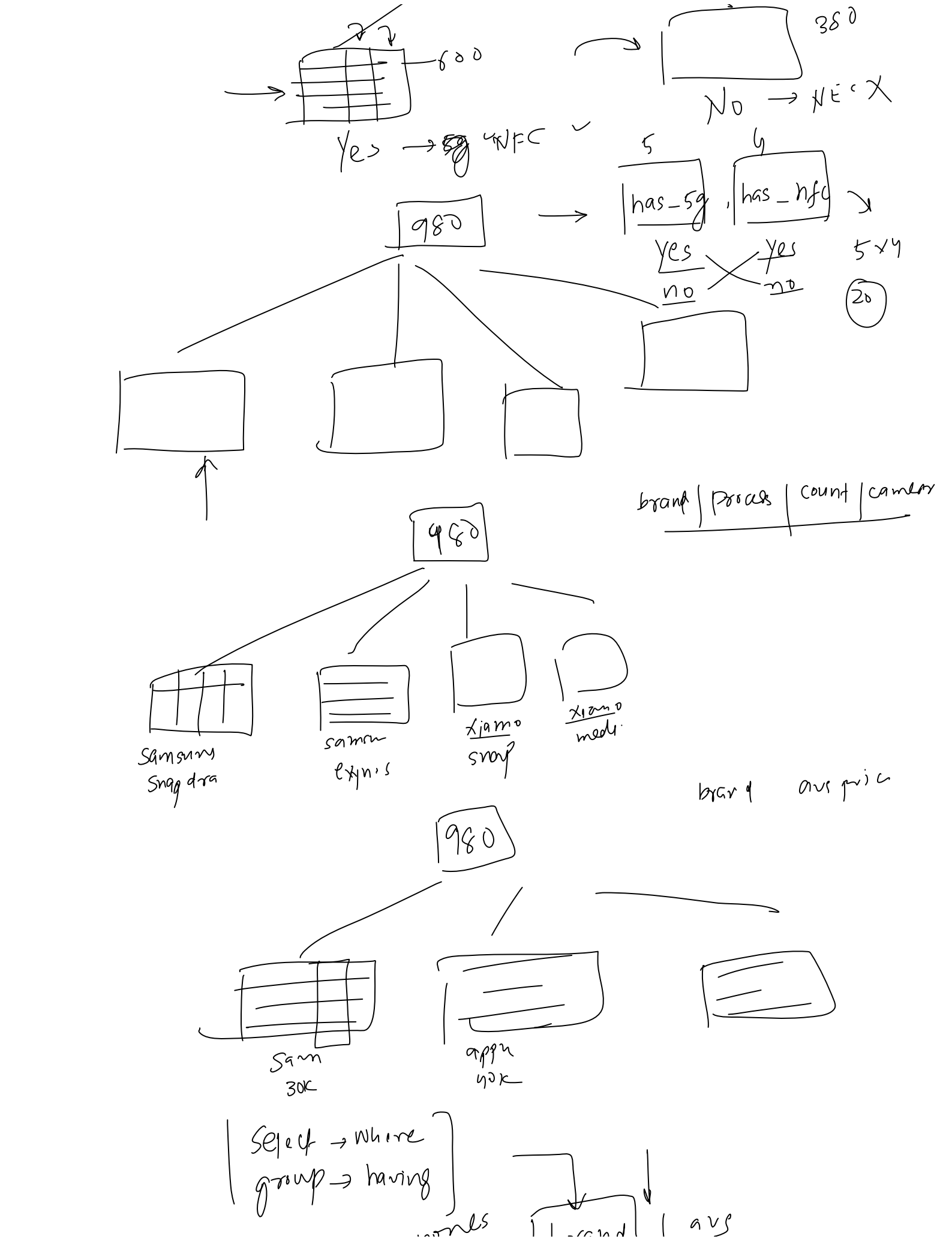
limit 10, 2 → 0-9 ✗ → 10, 11 ✓ 0-4 ✗

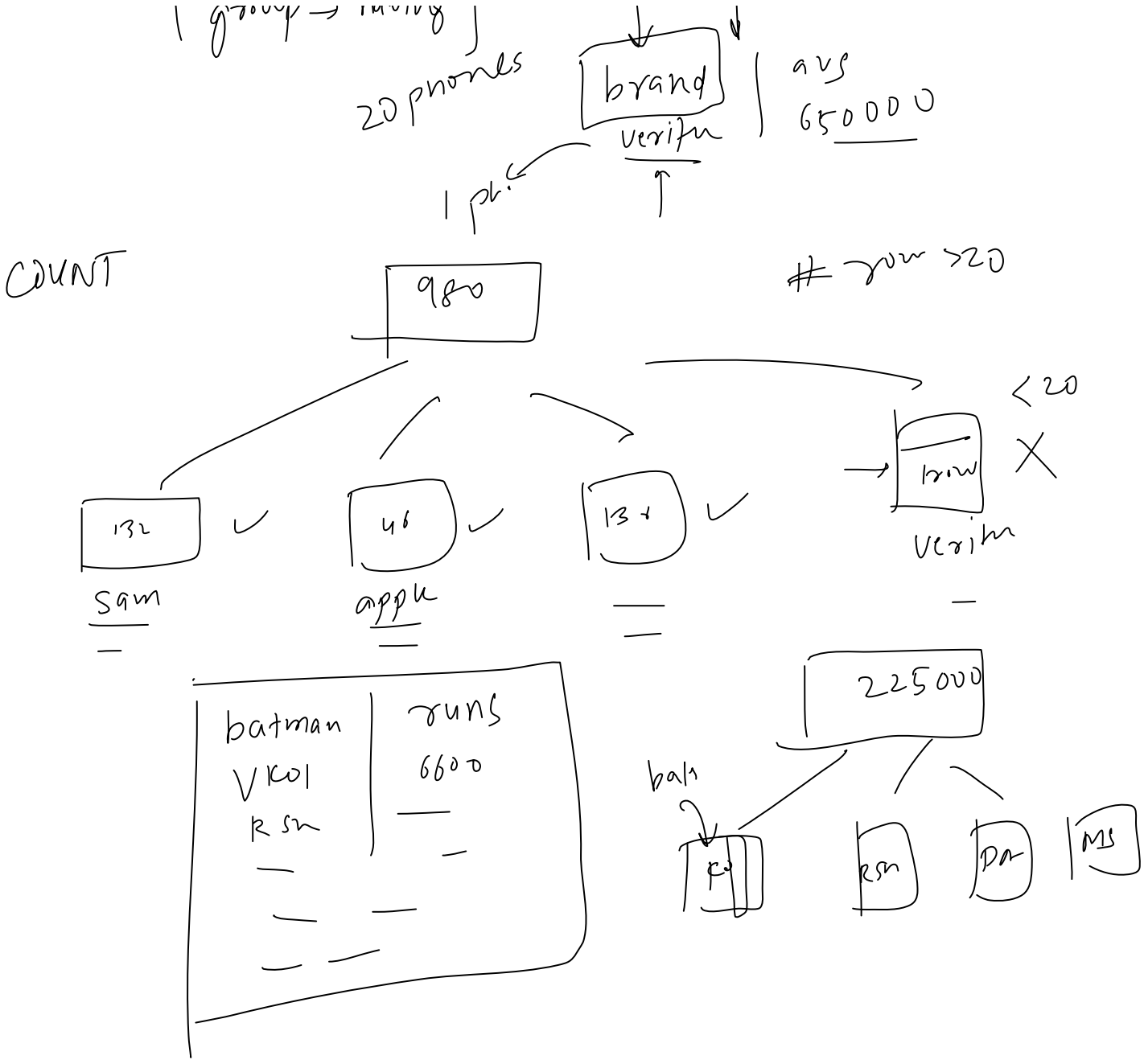
limit 5, 5 → 5, 6, 7, 8, 9

limit 1, 2 → 0 ✗ → 1 → 1, 2

→ groupby


980
groupby
brand name
num of phone
apple
samsung
xiaomi
Realme
41
41
132
47
980
sam
price 42
apple 34
samsun — 42
apple — 32
32K
62K
Table
aggregate
result
has_nfc
True
False
980
NFC
avg price
avg rating
yes
50000
81
no
30000
75
380


600
380
No → NFC X
Yes → 5g NFC
5
4
has_5g
has_nfc
yes
yes
no
no
5x4
20
980
brand | Prices | count | camera
480
Samsung
Snapdra
samsu
Exynos
Xiaomi
snap
Xiaomi
medi
brand avg price
980
Sam
30K
appl
40K
select → where
group → having
names
count
avg


1 group → having
20 phones
brand
avg
65000
verify
1 ph
COUNT
980
row >20
132
46
134
row
< 20
X
Sam
apple
Verify
batman
runs
Virat
6600
Rsh
225000
bats
Rsh
DA
MS

DDL commands for Databases

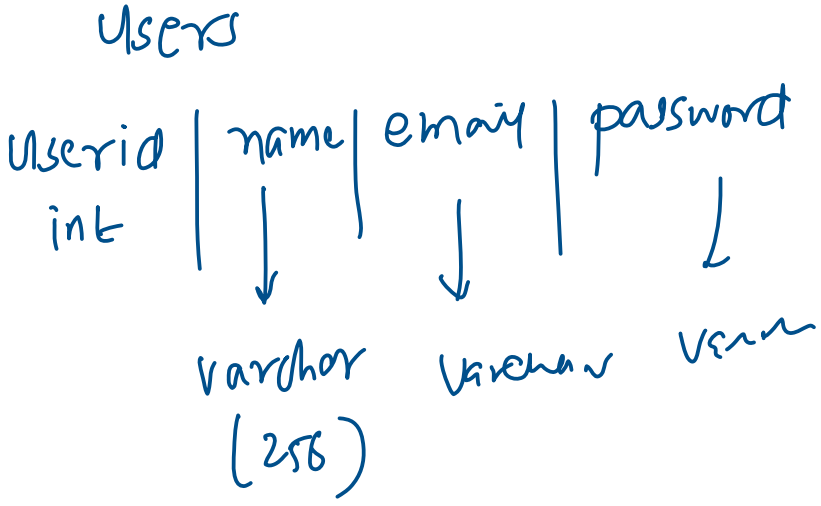
08 February 2023 14:57

1. CREATE → CREATE DATABASE campusx
2. DROP ↘ DROP DATABASE campusx

DDL commands for Tables

08 February 2023 15:00

1. CREATE
2. TRUNCATE
3. DROP

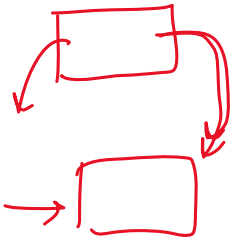


Data Integrity

08 February 2023 15:00

Data integrity in databases refers to the accuracy, completeness, and consistency of the data stored in a database. It is a measure of the reliability and trustworthiness of the data and ensures that the data in a database is protected from errors, corruption, or unauthorized changes.

There are various methods used to ensure data integrity, including:



Constraints:
Constraints in databases are rules or conditions that must be met for data to be inserted, updated, or deleted in a database table. They are used to enforce the integrity of the data stored in a database and to prevent data from becoming inconsistent or corrupted.

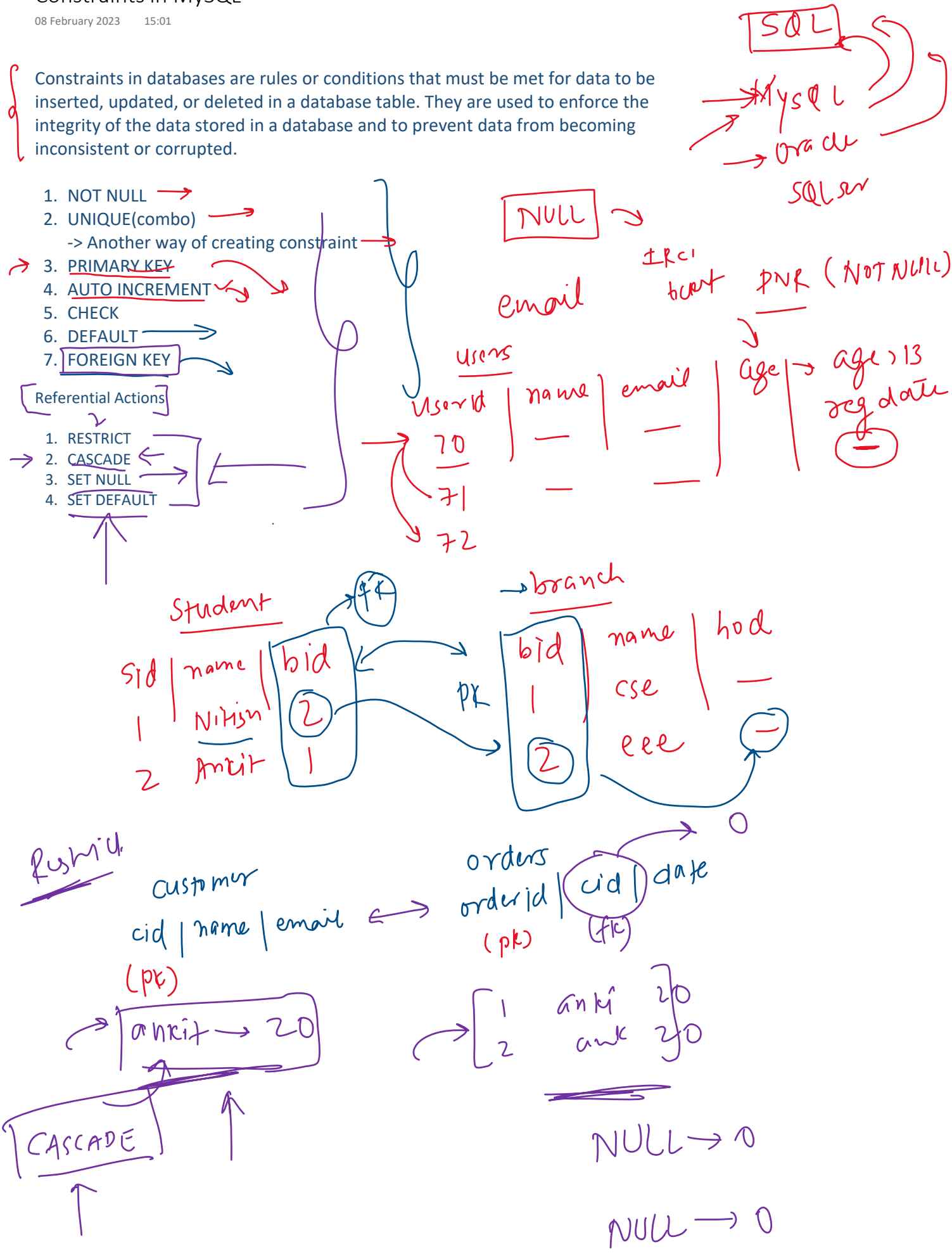
Transactions: a sequence of database operations that are treated as a single unit of work.

Normalization: a design technique that minimizes data redundancy and ensures data consistency by organizing data into separate tables.

Constraints in MySQL

08 February 2023 15:01

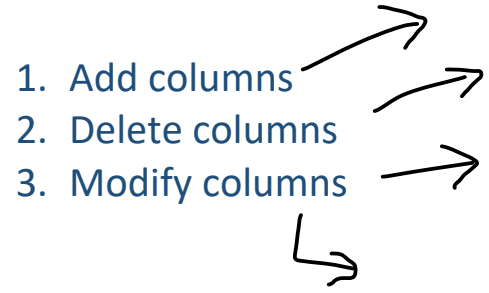
Constraints in databases are rules or conditions that must be met for data to be inserted, updated, or deleted in a database table. They are used to enforce the integrity of the data stored in a database and to prevent data from becoming inconsistent or corrupted.



ALTER TABLE command

08 February 2023 15:02

The "**ALTER TABLE**" statement in SQL is used to modify the structure of an existing table. Some of the things that can be done using the ALTER TABLE statement include



1. Add columns
2. Delete columns
3. Modify columns

Editing and Deleting Constraints

08 February 2023 15:02

1. Add
2. Delete
3. ~~Edit~~ nahi hota

What are SQL joins

20 February 2023 15:28

In SQL (Structured Query Language), a join is a way to combine data from two or more database tables based on a related column between them.

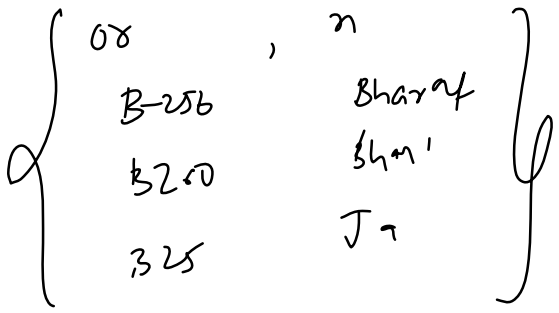
Joins are used when we want to query information that is distributed across multiple tables in a database, and the information we need is not contained in a single table. By joining tables together, we can create a virtual table that contains all of the information we need for our query.

But why have data in multiple tables?

20 February 2023 15:28

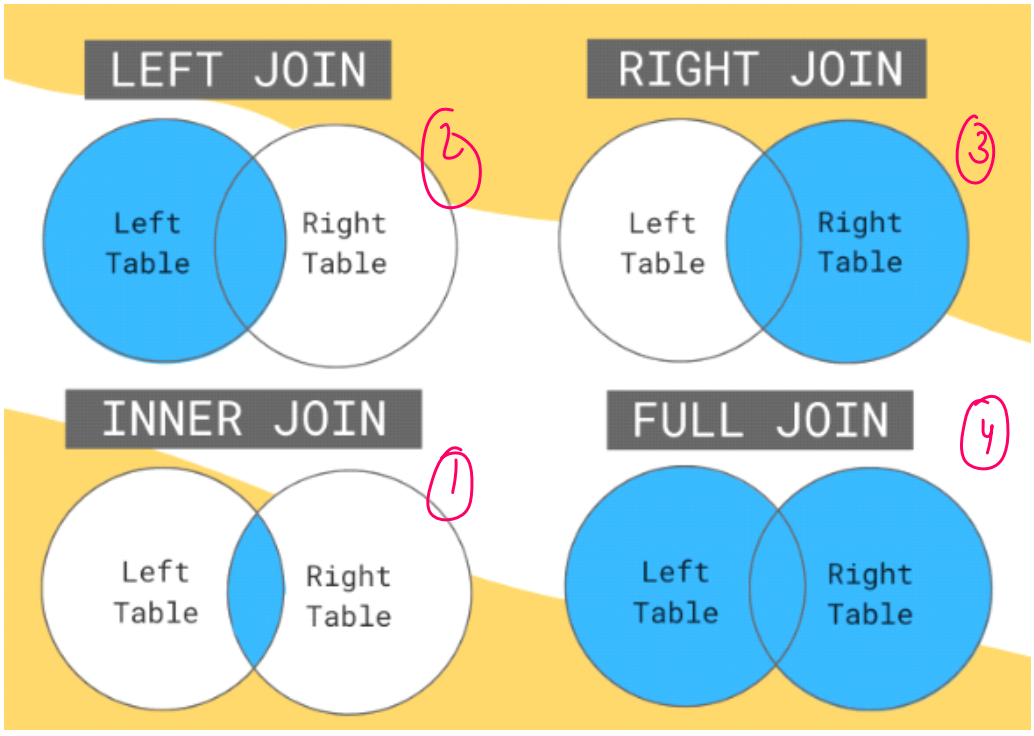
user_id	name	state	city
1	Bharat	Gujarat	Ahmedabad
2	Pearl	Maharashtra	Pune
3	Jahan	Madhya Pradesh	Bhopal
4	Divsha	Rajasthan	Jaipur
5	Kasheen	West Bengal	Kolkata
6	Hazel	Karnataka	Bangalore
7	Sonakshi	Jammu and Kashmir	Kashmir

order_id	user_id	order_date
B-25601	1	01-04-2018
B-26011	1	12-02-2019
B-26074	1	21-03-2019
B-25602	2	01-04-2018
B-26012	2	13-02-20
B-26075	2	21-03-2019
B-25603	3	03-04-2018
B-26013	3	13-02-2019



Types of Joins

20 February 2023 15:30



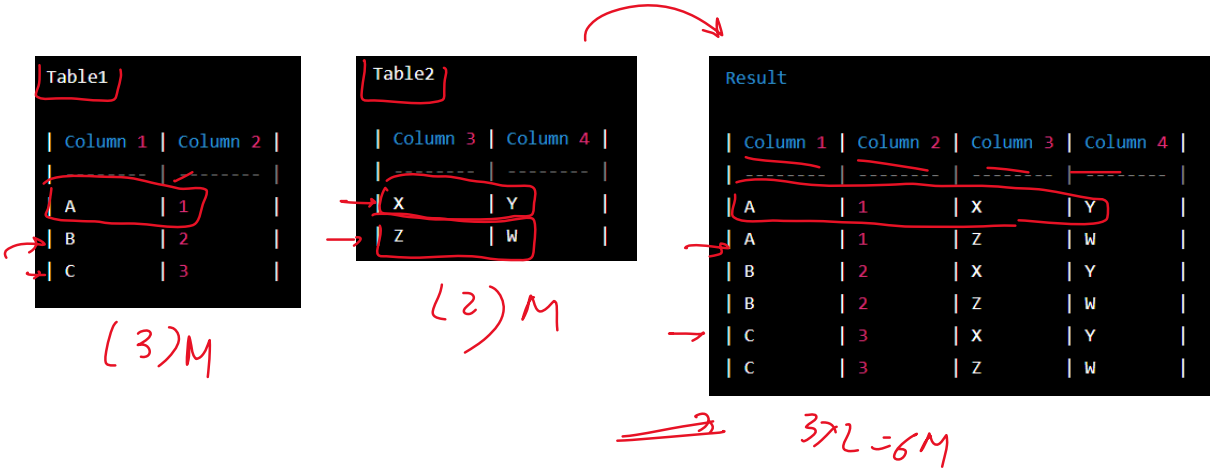
{ Cross join
self join }

Cross Joins -> Cartesian Products

20 February 2023 15:29

In SQL, a cross join (also known as a Cartesian product) is a type of join that returns the Cartesian product of the two tables being joined. In other words, it returns all possible combinations of rows from the two tables.

Cross joins are not commonly used in practice, but they can be useful in certain scenarios, such as generating test data or exploring all possible combinations of items in a product catalogue. However, it's important to be cautious when using cross joins with large tables, as they can generate a very large result set, which can be resource-intensive and slow to process.

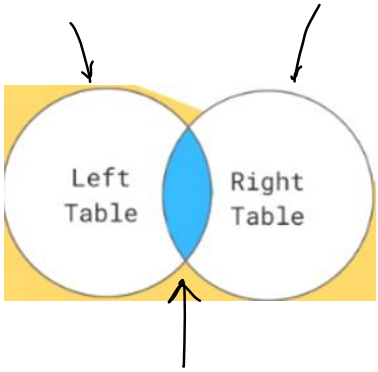


Inner Joins

20 February 2023 15:30

In SQL, an inner join is a type of join operation that combines data from two or more tables based on a specified condition. The inner join returns only the rows from both tables that satisfy the specified condition, i.e., the matching rows.

When you perform an inner join on two tables, the result set will only contain rows where there is a match between the joining columns in both tables. If there is no match, then the row will not be included in the result set.

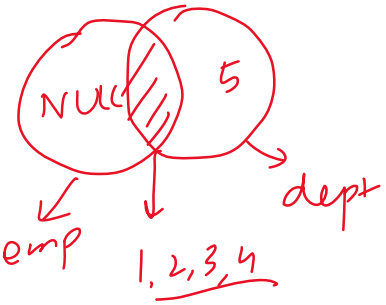


emp

Employee ID	Name	Department ID	Salary
1	John Smith	1	100000
2	Jane Doe	2	50000
3	Bob Johnson	3	75000
4	Lisa Wong	1	90000
5	Mike Lee	2	120000
6	Tim Davis	4	60000
7	Sarah Chen	NULL	80000

dept

Department ID	Department Name
1	Engineering
2	Sales
3	Finance
4	Marketing
5	Operations



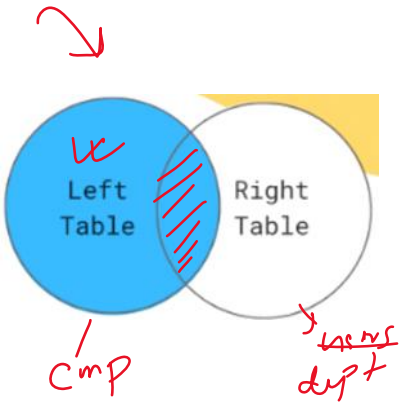
Employee ID	Name	Department ID	Salary	Department ID	Department
1	John Smith	1	100000	1	Engineering
4	Lisa Wong	1	90000	1	Engineering
2	Jane Doe	2	50000	2	Sales
5	Mike Lee	2	120000	2	Sales
3	Bob Johnson	3	75000	3	Finance

Left Join

20 February 2023 15:30

A left join, also known as a left outer join, is a type of SQL join operation that returns all the rows from the left table (also known as the "first" table) and matching rows from the right table (also known as the "second" table). If there are no matching rows in the right table, the result will contain NULL values in the columns that come from the right table.

In other words, a left join combines the rows from both tables based on a common column, but it also includes all the rows from the left table, even if there are no matches in the right table. This is useful when you want to include all the records from the first table, but only some records from the second table.



emp

Employee ID	Name	Department ID	Salary
1	John Smith	1	100000
2	Jane Doe	2	50000
3	Bob Johnson	3	75000
4	Lisa Wong	1	90000
5	Mike Lee	2	120000
6	Tim Davis	4	60000
7	Sarah Brown	5	80000
8	Mark Wilson	2	95000

dept

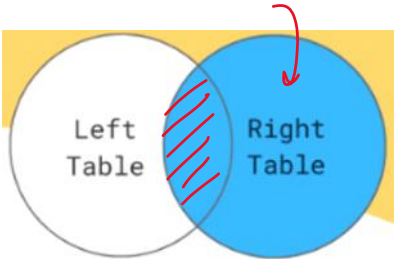
Department ID	Department Name
1	Engineering
2	Sales
3	Finance

Employee ID	Name	Department ID	Salary	Department ID	Department
1	John Smith	1	100000	1	Engineering
2	Jane Doe	2	50000	2	Sales
3	Bob Johnson	3	75000	3	Finance
4	Lisa Wong	1	90000	1	Engineering
5	Mike Lee	2	120000	2	Sales
6	Tim Davis	4	60000	NULL	NULL
7	Sarah Brown	5	80000	NULL	NULL
8	Mark Wilson	2	95000	2	Sales

Right Join

20 February 2023 15:31

A right join, also known as a right outer join, is a type of join operation in SQL that returns all the rows from the right table and matching rows from the left table. If there are no matches in the left table, the result will still contain all the rows from the right table, with NULL values for the columns from the left table.



emp

Employee ID	Name	Department ID	Salary
1	John Smith	1	100000
2	Jane Doe	2	50000
3	Bob Johnson	3	75000
4	Lisa Wong	1	90000
5	Mike Lee	2	120000
7	Sarah Brown	NULL	80000
8	Mark Wilson	2	95000

dept

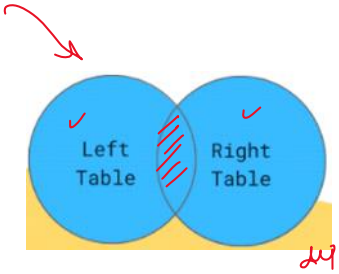
Department ID	Department Name
1	Engineering
2	Sales
3	Finance
4	Marketing
5	HR

Employee ID	Name	Department ID	Salary	Department Name
1	John Smith	1	100000	Engineering
4	Lisa Wong	1	90000	Engineering
2	Jane Doe	2	50000	Sales
5	Mike Lee	2	120000	Sales
8	Mark Wilson	2	95000	Sales
3	Bob Johnson	3	75000	Finance
NULL	NULL	4	NULL	Marketing
NULL	NULL	5	NULL	HR

Full Outer Join

20 February 2023 15:31

A full outer join, sometimes called a full join, is a type of join operation in SQL that returns all matching rows from both the left and right tables, as well as any non-matching rows from either table. In other words, a full outer join returns all the rows from both tables and matches rows with common values in the specified columns, and fills in NULL values for columns where there is no match.



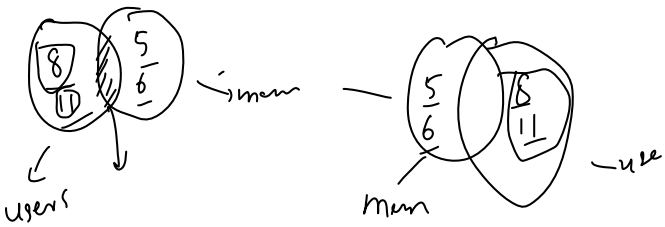
emp

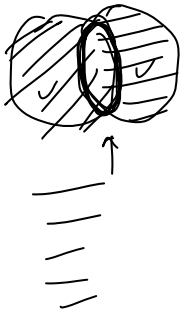
emp_id	emp_name	dept_id
1	Alice	1
2	Bob	1
3	Charlie	2
4	Dave	null
5	Eve	3

dept

dept_id	dept_name
1	Sales
2	Marketing
3	Finance
4	IT
5	HR

emp_id	emp_name	dept_id	dept_id	dept_name
1	Alice	1	1	Sales
2	Bob	1	1	Sales
3	Charlie	2	2	Marketing
4	Dave	null	null	null
5	Eve	3	3	Finance
null	null	null	4	IT
null	null	null	5	HR





SQL Set Operations

20 February 2023 15:35

1. **UNION**: The UNION operator is used to combine the results of two or more SELECT statements into a single result set. The UNION operator removes duplicate rows between the various SELECT statements.
2. **UNION ALL**: The UNION ALL operator is similar to the UNION operator, but it does not remove duplicate rows from the result set.
3. **INTERSECT**: The INTERSECT operator returns only the rows that appear in both result sets of two SELECT statements.
4. **EXCEPT**: The EXCEPT or MINUS operator returns only the distinct rows that appear in the first result set but not in the second result set of two SELECT statements.

person1 except person2

id	name
1	Alice
2	Bob
3	Charlie

id	name
3	Charlie
4	David
5	Emily

union

id	name
1	Alice
2	Bob
3	Charlie
4	David
5	Emily

union all

id	name
1	Alice
2	Bob
3	Charlie
3	Charlie
4	David
5	Emily

Intersect

id	name
3	Charlie

id	name
1	Alice
2	Bob

Self Joins

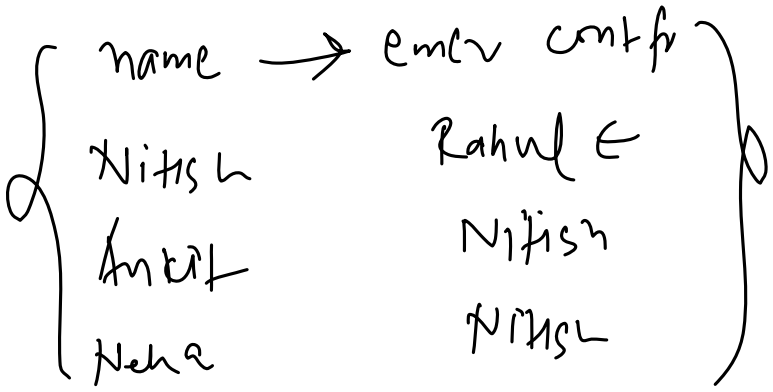
20 February 2023 15:34

A self join is a type of join in which a table is joined with itself. This means that the table is treated as two separate tables, with each row in the table being compared to every other row in the same table.

Self joins are used when you want to compare the values of two different rows within the same table. For example, you might use a self join to compare the salaries of two employees who work in the same department, or to find all pairs of customers who have the same billing address.

user_id	name	age	emergency_contact
1	Nitish	34	11
2	Ankit	32	1
3	Neha	23	1
4	Radhika	34	3
8	Abhinav	31	11
11	Rahul	29	8

user_id	name	age	emergency_contact
1	Nitish	34	11
2	Ankit	32	1
3	Neha	23	1
4	Radhika	34	3
8	Abhinav	31	11
11	Rahul	29	8



Joining on more than one cols

20 February 2023 16:58

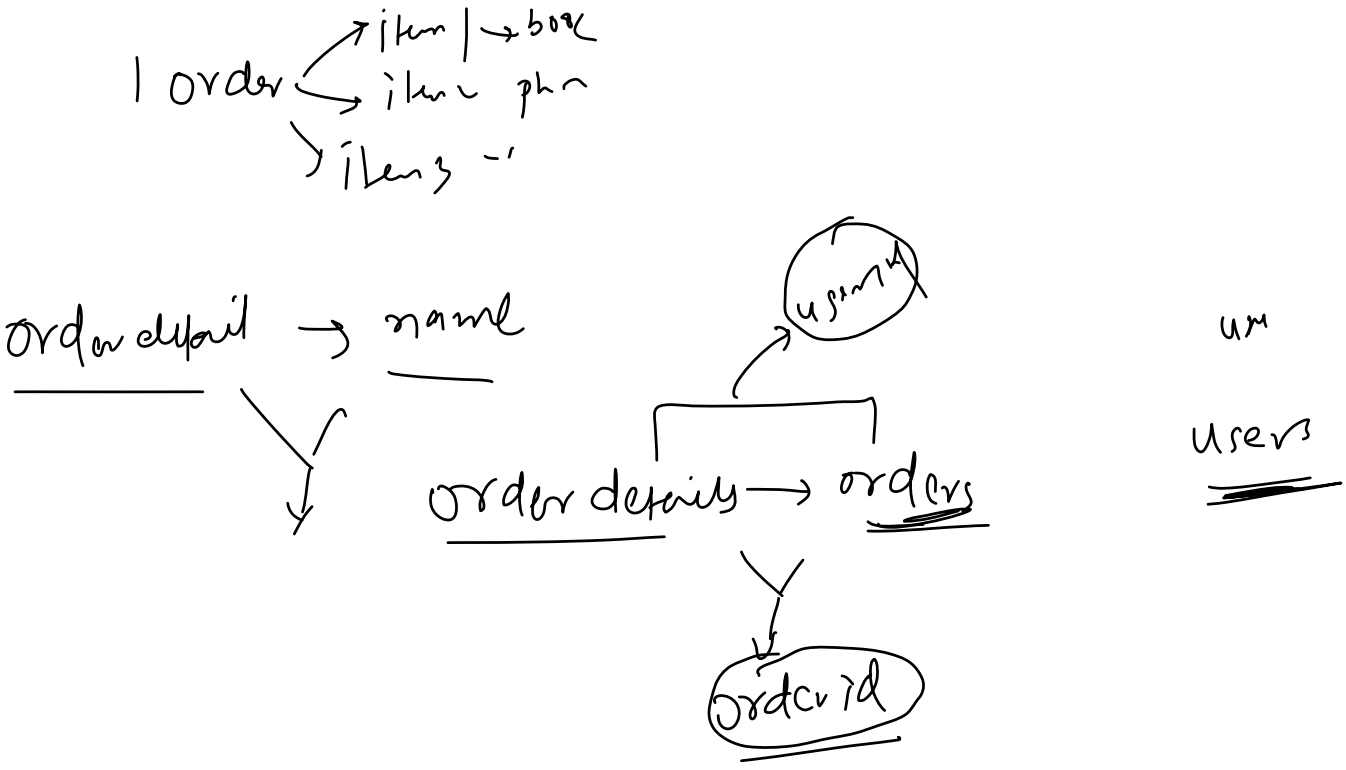
student_id	first_name	last_name	class_id	enrollment_year
1	John	Smith	1	2021
2	Jane	Doe	2	2020
3	Bob	Johnson	1	2021
4	Sally	Brown	3	2022
5	Tom	Williams	2	2022
6	Alice	Davis	4	2020

class_id	class_name	teacher	class_year
1	Math 101	Mr. Smith	2021
2	English 1	Ms. Johnson	2021
3	Science 1	Dr. Lee	2022
4	History 1	Ms. Williams	2022

Joining more than 2 tables

20 February 2023 15:33

1. Find order name and corresponding category name



Filtering Columns

20 February 2023 15:31

1. Find order_id, name and city by joining users and orders.
2. Find order_id, product category by joining order_details and category

Filtering Rows

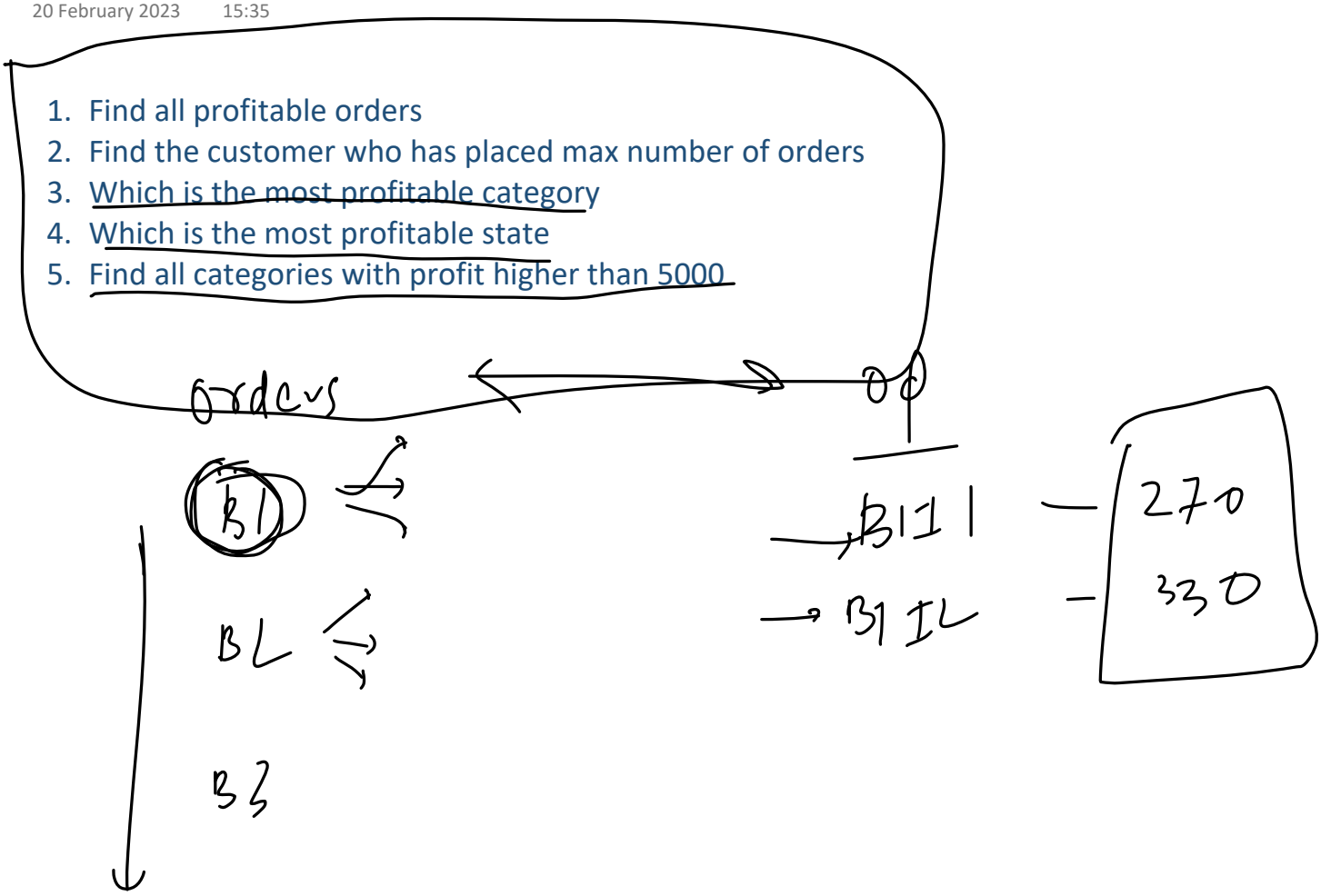
20 February 2023 15:32

1. Find all the orders placed in pune
2. Find all orders under Chairs category

Practice Questions

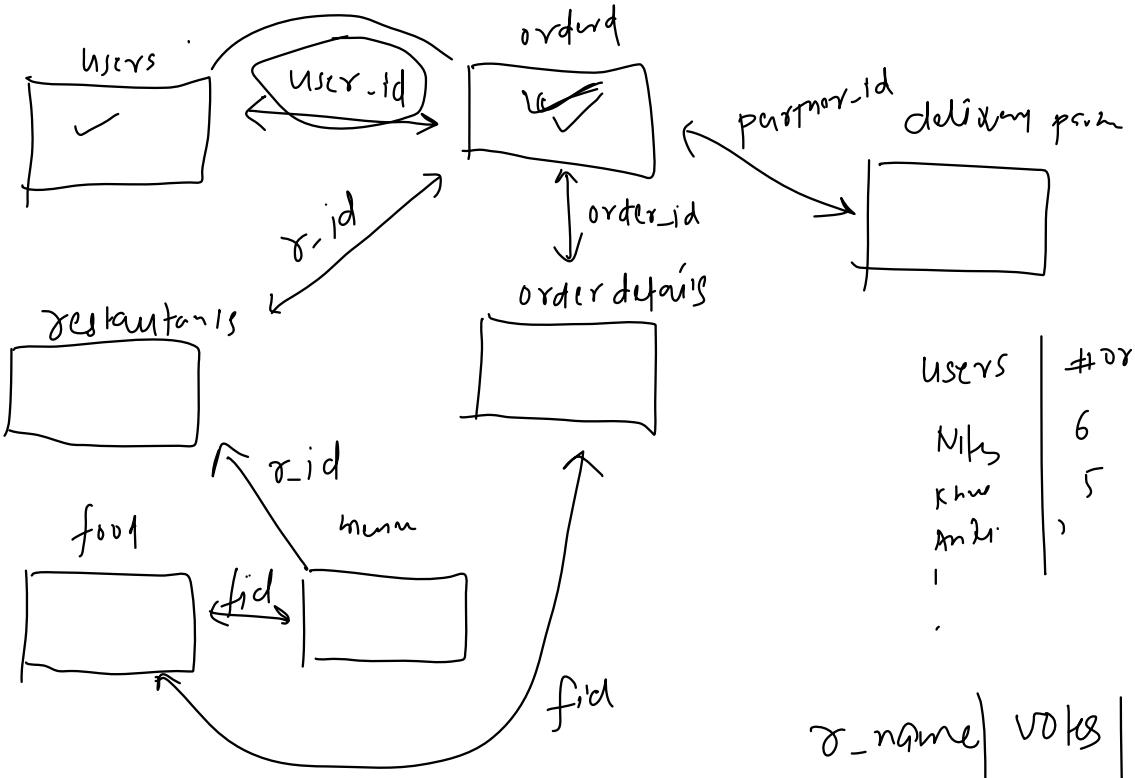
20 February 2023 15:35

1. Find all profitable orders
2. Find the customer who has placed max number of orders
3. Which is the most profitable category
4. Which is the most profitable state
5. Find all categories with profit higher than 5000



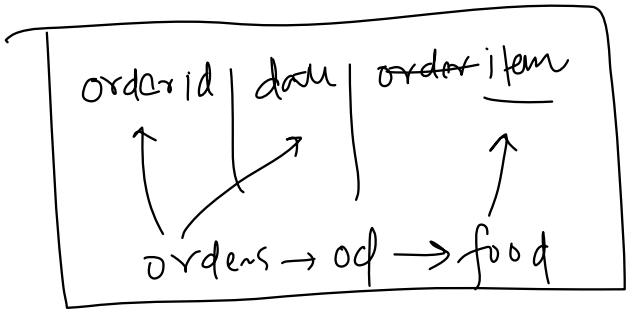
Zomato Case Study

21 February 2023 19:46



users	#order
Nik	6
Rahul	5
Amit	

r_name	votes	avg rating
~~Do~~ mic	500	3.9
ccc	600	4.1



15 may — 15 june

user	f_name	num
Nik	pizza	3
Rahul	Bir	4

name	salary

What is a Subquery

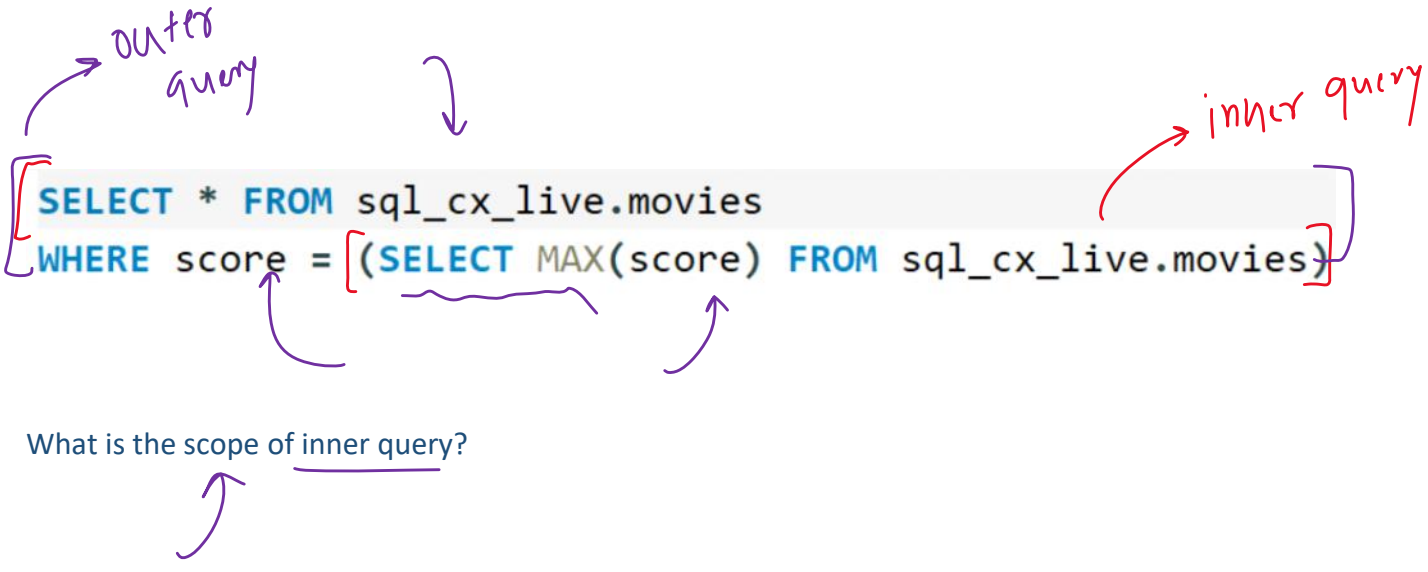
22 February 2023 14:25

In SQL, a subquery is a query within another query. It is a SELECT statement that is nested inside another SELECT, INSERT, UPDATE, or DELETE statement. The subquery is executed first, and its result is then used as a parameter or condition for the outer query.

Note - The topic is slightly difficult and needs a lot of practice

Example - Find the movie with highest rating

name	rating	genre	year	released	score	votes	director	writer
The Shining	R	Drama	1980	June 13, 1980 (United States)	8.4	927000.0	Stanley Kubrick	Stephen King
The Blue Lagoon	R	Adventure	1980	July 2, 1980 (United States)	5.8	65000.0	Randal Kleiser	Henry De Vere Stacpoole
Star Wars: Episode V - The Empire Strikes Back	PG	Action	1980	June 20, 1980 (United States)	8.7	1200000.0	Irvin Kershner	Leigh Brackett
Airplane!	PG	Comedy	1980	July 2, 1980 (United States)	7.7	221000.0	Jim Abrahams	Jim Abrahams
Caddyshack	R	Comedy	1980	July 25, 1980 (United States)	7.3	108000.0	Harold Ramis	Brian Doyle-Murray
Friday the 13th	R	Horror	1980	May 9, 1980 (United States)	6.4	123000.0	Sean S. Cunningham	Victor Miller
The Blues Brothers	R	Action	1980	June 20, 1980 (United States)	7.9	188000.0	John Landis	Dan Aykroyd
Raging Bull	R	Biography	1980	December 19, 1980 (United States)	8.2	330000.0	Martin Scorsese	Jake LaMotta
Superman II	PG	Action	1980	June 19, 1981 (United States)	6.8	101000.0	Richard Lester	Jerry Siegel
The Long Riders	R	Biography	1980	May 16, 1980 (United States)	7.0	10000.0	Walter Hill	Bill Bryden
Any Which Way You Can	PG	Action	1980	December 17, 1980 (United States)	6.1	18000.0	Buddy Van Horn	Stanford Sherman



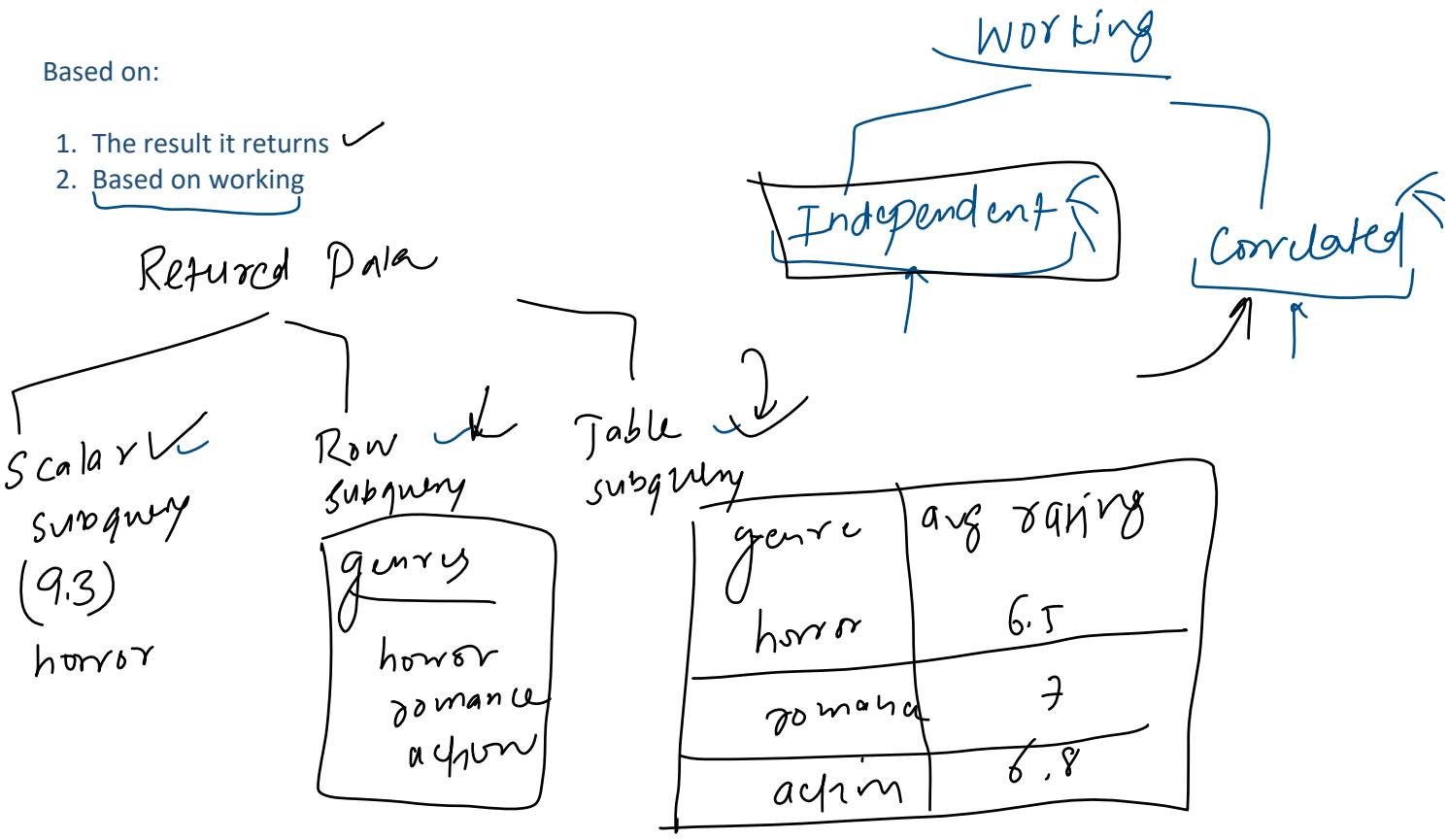
What is the scope of inner query?

Types of Subqueries

22 February 2023 14:26

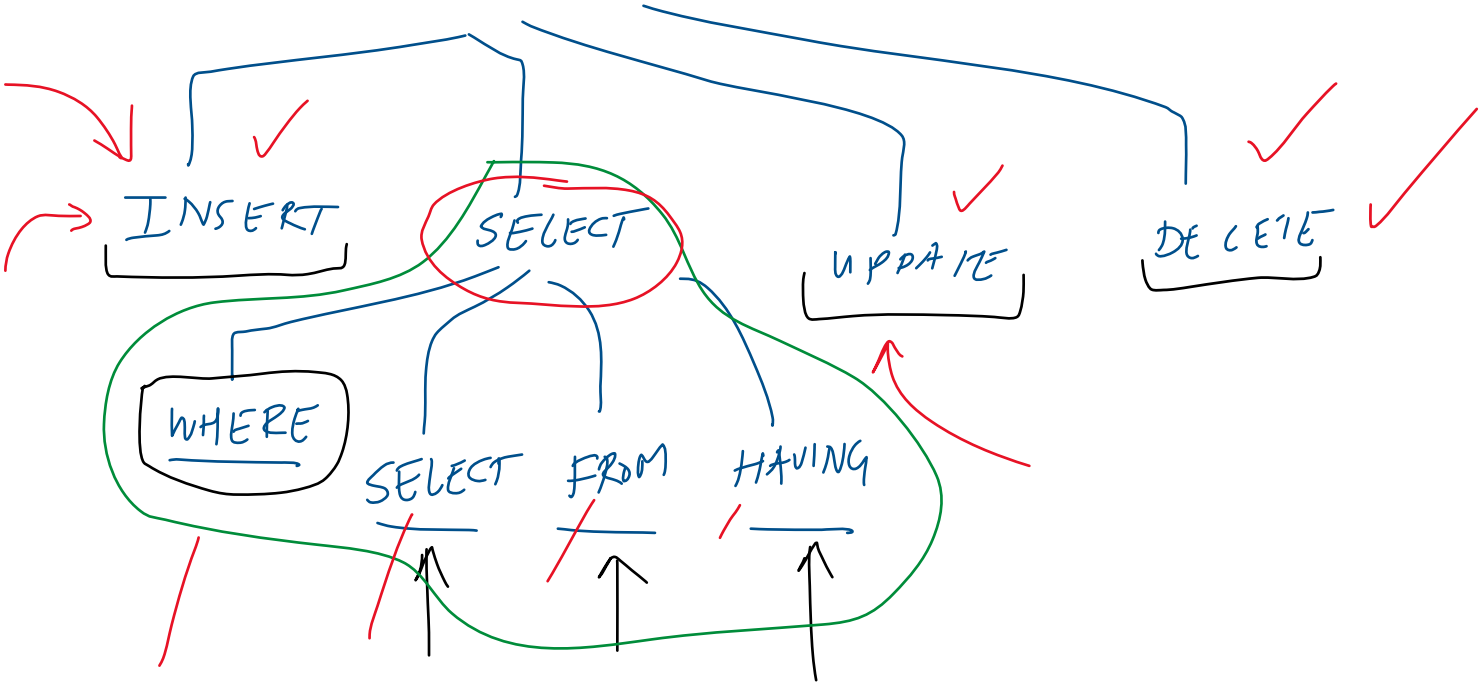
Based on:

1. The result it returns
2. Based on working



Where can subqueries be used?

22 February 2023 14:27



Independent Subquery - Scalar Subquery

22 February 2023 14:28

1. Find the movie with highest profit(vs order by)
2. Find how many movies have a rating > the avg of all the movie ratings(Find the count of above average movies)
3. Find the highest rated movie of 2000
4. Find the highest rated movie among all movies whose number of votes are > the dataset avg votes

O(n) + O(n) = O(2n)

O(n log n) + O(1)

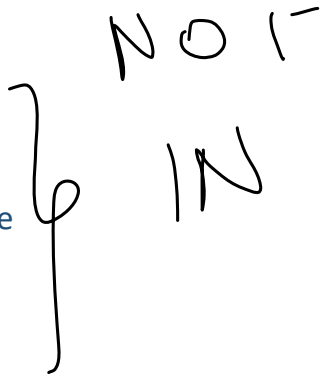
O(n log n)

index - x

Independent Subquery - Row Subquery(One Col Multi Rows)

22 February 2023 14:29

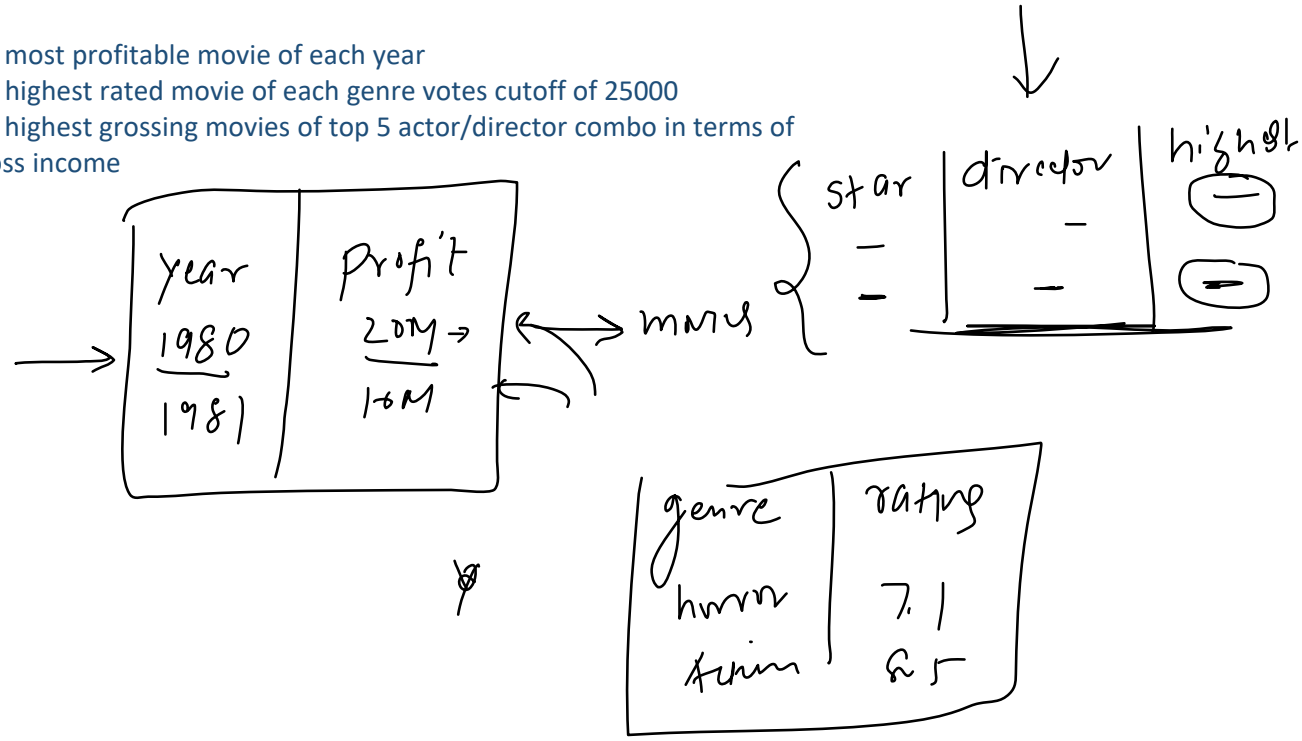
1. Find all users who never ordered
2. Find all the movies made by top 3 directors(in terms of total gross income)
3. Find all movies of all those actors whose filmography's avg rating > 8.5(take 25000 votes as cutoff)



Independent Subquery - Table Subquery(Multi Col Multi Row)

22 February 2023 14:29

1. Find the most profitable movie of each year
2. Find the highest rated movie of each genre votes cutoff of 25000
3. Find the highest grossing movies of top 5 actor/director combo in terms of total gross income



Correlated Subquery

22 February 2023 14:29

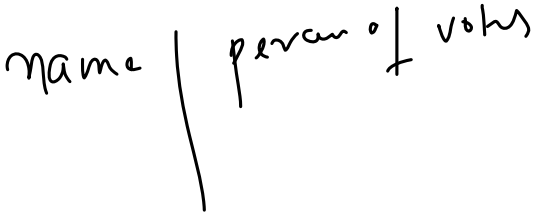
1. Find all the movies that have a rating higher than the average rating of movies in the same genre.[Animation]
2. Find the favorite food of each customer.

Usage with SELECT

22 February 2023 14:29

1. Get the percentage of votes for each movie compared to the total number of votes.
2. Display all movie names ,genre, score and avg(score) of genre

-> Why this is inefficient?



Usage with FROM

22 February 2023 14:30

1. Display average rating of all the restaurants

Usage with HAVING

22 February 2023 14:30

1. Find genres having avg score > avg score of all the movies

Subquery In INSERT

22 February 2023 14:30

Populate a already created loyal_customers table with records of only those customers who have ordered food more than 3 times.

Subquery in UPDATE

22 February 2023 14:31

Populate the money col of loyal_cutomer table using the orders table. Provide a 10% app money to all customers based on their order value.

Subquery in DELETE

22 February 2023 14:31

Delete all the customers record who have never ordered.